ThreatReady: Comprehensive UI/UX Audit

Repository	`Ricky-Hacker001/Threat_Ready`, branch `main`, commit `93cc4bf`
Audit date	23 October 2026
Type	Audit only. No application source, styles, configuration or dependencies were changed. This document and the screenshots in `docs/audit-evidence/` are the only additions to the repository.
Covers	UI/UX, overflow, usability and cognitive load, learning and simulation workflows, responsive layout, accessibility, functional UI, visual consistency, browser console, and a prioritised fix plan
Status	**Fixes applied** on local branch `ui-ux-audit-fixes` (not committed, not pushed). 86 of 96 findings are fixed, 6 are partly fixed and 4 are left open on purpose. See §17 Fix status. Sections 1–16 are the original audit, kept as the baseline.

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1. Executive summary

Verdict

ThreatReady's content is strong: realistic scenarios, honest "simulated environment" messaging, and feedback that explains why an answer was right or wrong. The SOC after-action review and the ThreatWall verification list are the best examples. The interface around that content is inconsistent, though. Ten apps have grown side by side, and each has its own conventions for layout, colour, keyboard support and saving progress.

Four groups of problems matter most:

1. **Broken or misleading journeys.**
    - The marketing site's sign-up and sign-in buttons point to a path that does not exist.
    - The Blue Team track says it has no labs, but the SOC lab is live.
    - The free trial promises usage that it then blocks.
    - In the AI Security Lab, an attempt that fails its re-test cannot be finished, and a page reload loses the attempt.
2. **Layout that breaks inside the platform.**
    - When a lab runs inside the platform, a thin bar (the lab-nav strip) is added to the top of the page. That bar changes how the page height is calculated.
    - In TR-CLOUD-001 at 1366×768, the console grows to 1,488 px tall and the action dock starts at y = 1,318, below the fold. *Confirmed in a browser.*
    - Ten pages scroll sideways on phones. *Confirmed in a browser.*
3. **Accessibility gaps on core tasks.**
    - Keyboard focus gets stuck in two places: the DevSecOps code editor and the Red Team terminal. *Confirmed*.
    - The Red Team lab has almost no accessibility markup (one ARIA attribute in the whole client).
    - The AI Security architecture map is scored, but it cannot be used with a keyboard.
    - The brand violet `#6D5DFC` is used as text on dark backgrounds at 3.9–4.2:1, below the 4.5:1 minimum, in at least five apps.
4. **No shared design system in practice.**
    - There are three button systems, two container widths and four different approaches to light theme. Four lab apps ignore the user's theme choice altogether.
    - Errors in InterviewPrep use the same violet as primary buttons.
    - The theme does not carry over between threatready.io and the signed-in apps.

Numbers

Measure	Result
Apps and modules found	10 deployable front ends + 1 deployment gateway + 1 cross-browser e2e harness
Pages and states tested in a browser	36 (×8 viewports = 288 layout checks), plus 5 multi-step workflows at 3 viewports
Layout checks that failed (horizontal page overflow)	**19 of 288**, across **10 pages/states** (plus 1 more found in a multi-step walkthrough: AI Security mission control)
Automated accessibility violations (axe-core, WCAG 2.1 A/AA + best practice)	11 rule types; **241 colour-contrast nodes** across 16 of 36 pages
Findings in this report	**96** (Critical 3 · High 21 · Medium 43 · Low 29)
Apps blocked for runtime testing	3: onboard, InterviewPrep, Attack Reasoning (they need PostgreSQL and LLM services, and Docker is not installed on this machine)

Top 12 issues to fix first

#	ID	Severity	Issue	Evidence
1	FUNC-01	Critical	About 74 marketing CTAs ("Sign in", "Start free") point to `threatready.io/app/…`, which does not exist	Source confirmed
2	FUNC-02	Critical	When hosted, the SOC live log stream and downloads use absolute `/api/…` paths, so they miss the SOC app	Source confirmed; prior audit confirmed it in the built bundle
3	OVF-01	Critical	The lab-nav strip forces `height:auto`. In the TR-CLOUD-001 console the dock drops below the fold and the panes stop scrolling on their own. ThreatWall and DevSecOps become 40 px taller than the screen	**Runtime confirmed**
4	UX-01	High	The Blue Team track shows "No labs in this track yet" while the SOC lab is live	**Runtime confirmed**
5	FUNC-04 / FUNC-05	High	AI Security Lab: a failed re-test cannot reach results; refresh, Back or Exit loses the attempt	Runtime confirmed (reload); source confirmed (finish)
6	A11Y-01	High	Keyboard focus is trapped in the DevSecOps editor and the Red Team terminal	**Runtime confirmed**
7	A11Y-02	High	Red Team lab: unlabelled inputs, mouse-only controls, modals that aren't marked as dialogs, toasts that screen readers never announce, and hidden drawers that keyboard focus can still reach	Source confirmed + runtime (drawers)
8	A11Y-05	High	Violet `#6D5DFC` used as text on dark backgrounds fails 4.5:1 (SOC, DevSecOps, AI Security, onboard links, rubric page)	**axe, runtime confirmed**
9	VIS-01	High	Attack Reasoning score-breakdown bars render nothing, because no CSS rule styles their markup	Source confirmed
10	FUNC-03	High	Free trial promises "1 interview, 2 simulations and 1 reasoning session", but every product then says "Needs a free account"	Source confirmed
11	CONT-01	High	Legal pages still contain the placeholders "[Registered Office Address]" and "[To be appointed]"	Source confirmed
12	UX-07	High	InterviewPrep voice answers auto-submit after about 9.5 s of silence, and nothing announces the countdown	Source confirmed

2. Scope, method and limitations

2.1 Evidence labels

Every finding carries one of these labels:

Label	Meaning
RC: runtime confirmed	Reproduced in a real browser (Google Chrome, driven by `playwright-core` 1.53.2). Measurements come from DOM geometry or from axe-core 4.10.3.
SC: source confirmed	The code or CSS clearly produces the behaviour. Not observed in a browser, usually because the app could not run here.
SI: source-inferred risk	Likely from reading the code, but needs a browser check before anyone fixes it.
PA: prior audit	Taken from `docs/UX_AUDIT.md` (14 Oct) or `docs/AUDIT_REPORT_2026-10-15.md` and re-checked against the current source.

2.2 Severity scale

Severity	Definition
Critical	Blocks a primary journey for most users, loses work or data, or makes a main screen unusable at a common size
High	A core task fails or is seriously degraded; a WCAG 2.1 A/AA failure on a core flow; a confirmed layout break at a supported viewport; misleading product state
Medium	Slows people down or causes confusion, but a workaround exists; a WCAG issue on a secondary flow; design-system drift that users can see
Low	Polish, consistency or wording

2.3 What was done

1. **Repository discovery.** I read the root `README.md`, every app's README and docs, the package manifests, the routers, the shared CSS, `deploy/` and the e2e harness. I also read the two existing audits.
2. **Source review.** Five parallel read-only reviews covered every front end, with file:line evidence:
    - onboard + marketing site
    - InterviewPrep

- Labs hub + Attack Reasoning
- SOC + Red Team
- ThreatWall + DevSecOps + AI Security

I also ran repository-wide pattern counts (viewport units, `nowrap`, media queries, raw hex colours, ARIA, reduced motion).

3. **Runtime testing.** Every app that can run without PostgreSQL ran locally from its own dev command, using each app's built-in development sign-in where one exists.

   - The Labs hub needs onboard's identity service. For it, I used a small local stub of the internal endpoint `GET /api/internal/identity/session`, which returns one fixed test user. The stub lived in a temporary folder outside the repository.
   - An automated sweep then measured each page at the 8 target viewports. It recorded horizontal overflow and the elements causing it, touch targets under 24 px, `h1` count, `lang`, console errors and failed requests. It also ran axe-core at 1440×900 (and at 375×812 on entry pages).
   - Scripted walkthroughs drove five multi-step workflows at 375, 768 and 1366 px.
   - Targeted DOM checks covered keyboard traps, off-screen focusable drawers, editor clipping, reload persistence, unknown routes, the lab-nav height behaviour and the light theme.

4. **Verification of the prior UX audit.** See §13.

2.4 Coverage

App	Source review	Runtime	How it ran	Not covered
`Landing_Page-main` (threatready.io)	Full	**All 17 pages × 8 viewports + axe**	Static server	Real CDN headers; the GitHub Pages 404 rewrite
`Threatready-landing-page-main` (onboard)	Full	**BLOCKED**	Needs PostgreSQL, Better Auth, email and Turnstile	All runtime behaviour
`InterviewPrep`	Full	**BLOCKED**	Needs PostgreSQL, the pg-boss worker and an Anthropic key	All runtime behaviour
`threatready-lab-001` (Labs catalog + TR-CLOUD-001)	Full	Catalog, 4 tracks, an unknown track and the console entry, × 8 viewports; console layout checks; light theme	Labs API with the local identity stub	Full TR-CLOUD-001 playthrough, report submission
`attack-reasoning`	Full	**BLOCKED**	Needs the InterviewPrep API, PostgreSQL and the LLM worker	Attempt, evaluation and result pages

App	Source review	Runtime	How it ran	Not covered
`soc`	Full	Home, level, demo intro, badges, case console (monitor stage)	`npm run start:dev` (DEV_USER)	Hosted `/labs/soc/` path, L2/L3 stages, report
`red_team_lab/web`	Full	Login, catalog, briefing, all 6 tool tabs at 3 viewports, light theme	Built-in dev login	Exploit-to-report playthrough
`ThreatWall`	Full	Home, lab entry, notebook, record × 8 viewports; lab-nav simulation	`vite`	Tasks 0.1–4.x playthrough
`DevSecOps-Simulation`	Full	Welcome, mode choice, dashboard, lab 0 × 8 viewports; keyboard checks; lab-nav simulation	`vite`	Labs 1–7
`AI-Security`	Full	Entry, mission control, briefing, workspace; reload test; lab-nav simulation	`vite`, standalone (name entry, because the platform identity is absent)	Re-test, results, report

Not covered anywhere:

- the single-origin Render deployment (`deploy/gateway.mjs`)
- Firefox and WebKit
- screen readers (NVDA, VoiceOver, TalkBack)
- physical touch devices and virtual keyboards
- browser zoom at 200% and 400%
- complete end-to-end scenario playthroughs

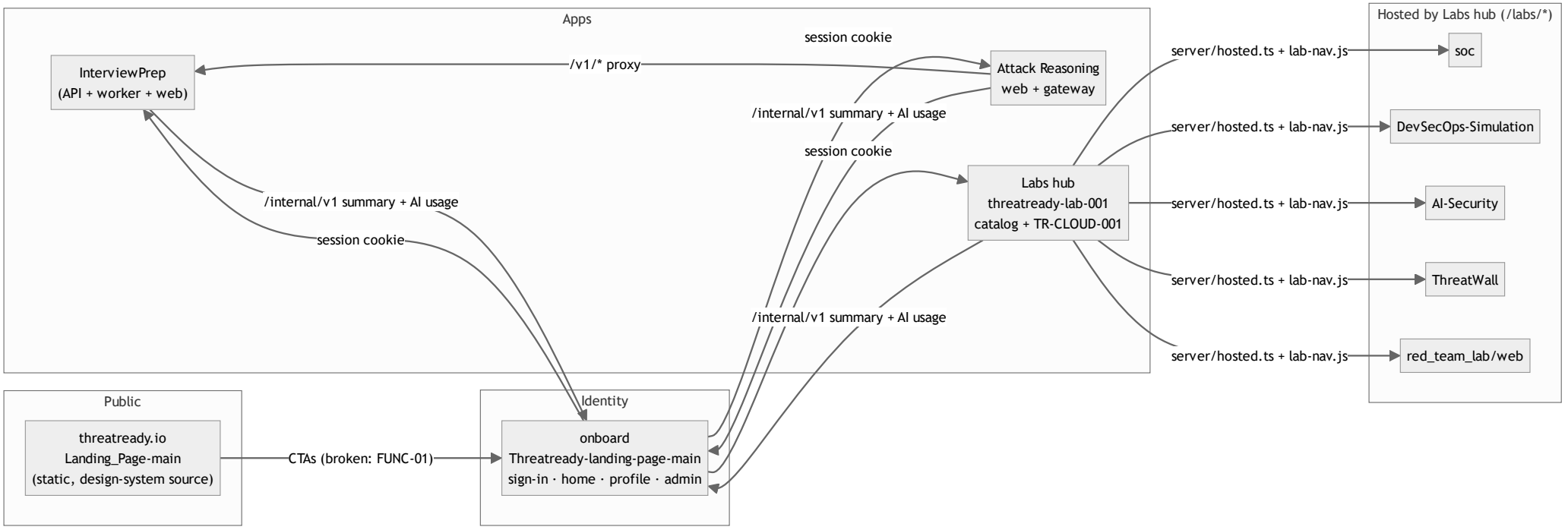
A route marked **PASS** in §9 means only that the page rendered and did not scroll sideways at that size. It does **not** mean a full visual or functional sign-off.

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3. Repository discovery and architecture

3.1 What the repository is

One monorepo folder holds ten front ends. They share one sign-in (onboard, built on Better Auth) and one session cookie. In the free deployment (`render.yaml`, `deploy/`), they are all served from a single origin by one Node gateway. The root README calls this design "One sign-in, separate apps".



3.2 Inventory

Directory	Purpose (from source)	Entry points	Tech	Depends on	Runtime status	Uncertainty
`Landing_Page-main/`	Public marketing site; `assets/css/site.css` is called the design-system source	17 `*.html` files, `assets/js/site.js`	Static HTML/CSS/JS	none (CTAs should point to onboard)	Tested	"Lab Simulations: in development" (41 places) contradicts the live labs

Directory	Purpose (from source)	Entry points	Tech	Depends on	Runtime status	Uncertainty
`Threatready-landing-page-main/`	Identity provider, signed-in home, profile, admin console	`apps/web/src/main.tsx`, `apps/api`	React 19, RR7, Better Auth, NestJS-style API, pnpm	PostgreSQL, Resend, Turnstile, each app's `/internal`	Blocked	Folder name suggests a landing page, but it is the **onboard** app
`InterviewPrep/`	AI mock interviews; also hosts the **reasoning engine** for Attack Reasoning	`apps/web/src/main.tsx`, `apps/api/src/main.ts`	React 19, TanStack Query, three.js, Vite 8	PostgreSQL, pg-boss, Anthropic, onboard identity	Blocked	`better-auth` still a dependency though unused
`threatready-lab-001/`	Labs catalog + TR-CLOUD-001 AWS-style console; **hosts the five other labs**	`client/src/main.tsx`, `server/index.ts`, `server/hosted.ts`	React, Express 5, JSON file store	onboard identity	Tested with stub	Name suggests one lab, but it is the Labs hub
`attack-reasoning/`	Attack Reasoning Lab SPA + thin gateway	`client/src/main.tsx`, `server/index.ts`	React, @xyflow/react, dagre	InterviewPrep API (`/v1`), onboard	Blocked	`/scenarios` pages are orphaned legacy
`soc/`	SOC simulation: live logs, L1→L3 ticket, after-action review	`client/src/main.jsx`, `server/index.js`	React 18 JSX, Express 4	Platform user when hosted	Tested	`soc/render.yaml` looks stale
`red_team_lab/web/`	TR-RED-001 web pentest lab	`client/src/main.tsx`, `server/index.ts`	React, xterm.js, Express	Platform user when hosted	Tested	Its `platform.css` copy has drifted
`ThreatWall/`	Firewall administration lab, levels 0–4	`src/main.tsx`	React 19, Vite 8	Browser only (localStorage)	Tested	Deliberately looks like a vendor appliance
`DevSecOps-Simulation/`	Scanners, fixes and a gated pipeline in the browser	`src/main.tsx`	React 18, Vite 6	Browser only	Tested	No tests
`AI-Security/`	30 LLM attack-detect-fix scenarios	`src/main.tsx`	React 18, Vite 5	`/labs/api/me` for the user	Tested	README says "Nova"; status doc says "Aero"
`deploy/`	Single-service Render launcher and gateway	`start.mjs`, `gateway.mjs`, `build.sh`	Node	everything	Not run	—
`e2e/`	Cross-browser journeys (Chromium, Firefox, WebKit)	`cross-browser.mjs`	Playwright	the whole local stack plus a test account	Not run (needs the full stack)	Checks overflow at 2 widths only (§12)
`docs/`	Prior UX audit, security audit, AR design	—	—	—	—	—

Duplicates and orphans found:

- `platform.css` is byte-identical in onboard, InterviewPrep, Labs and Attack Reasoning, but the **Red Team copy has drifted** (257 vs 295 lines; it lacks `.tr-loading` and view transitions).

- `shell.tsx` is identical in Labs and Attack Reasoning. Onboard uses its own `components/Platform.tsx`.
- Orphaned code:
  - Attack Reasoning `/scenarios` and `/scenarios/:id`: no links lead there; the copy is stale; "Start scenario" has no handler.
  - Labs `GET /labs/:id/runs`: never called.
  - Red Team `api.catalog()`: never called.
  - ThreatWall `PlannedPage`: dead code.
  - onboard `api.feature` and `api.launch`; plus dead CSS (`.topbar`, `.profile-menu*`).
  - InterviewPrep `.auth*` and `.skip-link` CSS.
- `/interview-prep/__capture` is a development page with no auth guard (`InterviewPrep/apps/web/src/app/App.tsx:109`).

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4. Route, page and feature inventory

"Rendered" means the page was loaded in a browser during this audit.

4.1 Public pages: `Landing_Page-main`

Page	Purpose	Rendered	Findings
`index.html`	Home, product tabs, demo, FAQ	Yes, 8 viewports	FUNC-01, FUNC-03 (FAQ trial copy), CONT-03
`attack-reasoning-lab.html`	Product page	Yes	A11Y-11 (breadcrumb link contrast)
`lab-simulations.html`	Product page, titled "(in development)"	Yes	CONT-03
`interviewprep.html`	Product page	Yes	A11Y-11
`pilot.html`	Pilot request form	Yes	**OVF-09**
`pricing.html`	Plans; "entitlements are being confirmed"	Yes	CONT-01
`transparent-scoring.html`, `rubric.html`	Scoring explanations (overlap)	Yes	A11Y-05 (rubric: 33 contrast nodes)
`how-it-works.html`, `resources.html`, `contact.html`	Info	Yes	A11Y-11; the footer "Contact" opens a new tab
`sample-report.html`	Sample skill report (own inline CSS)	Yes	**OVF-10**, VIS-06, A11Y-05 (35 nodes)
`security.html`, `privacy.html`, `terms.html`, `dpa.html`	Legal and trust	Yes	CONT-01, A11Y-12
`404.html`	404 page + `/app/*` redirect script	Yes	FUNC-01 (possible redirect loop)

4.2 Authentication and account: onboard (`Threatready-landing-page-main/apps/web/src/main.tsx:21-46`)

Route	Component	Auth	Purpose	Rendered
`/`	`AuthPage`	Public	Sign in, sign up, verify, forgot username/password, trial, OAuth	Blocked
`/signout`	`SignOutPage`	Any	Single sign-out	Blocked
`/app`	`ProductPage`	Session	Greeting, "Continue" strip, products	Blocked
`/app/profile`	`ProfilePage`	Session	Cross-app overview, account details	Blocked

Route	Component	Auth	Purpose	Rendered
`/app/admin`	`AdminPage`	Admin	Users, sign-ups, AI usage and cost	Blocked
`/app/admin/users/:id`	`AdminUserPage`	Admin	Per-user detail, role change	Blocked
`/app/:feature`	`ModulePage`	Session	Legacy redirect to an app subdomain	Blocked

4.3 Dashboards and learning apps

App	Route	Purpose	Rendered
InterviewPrep	`/interview-prep`	Dashboard	Blocked
	`/interview-prep/new`	Resume + job description + consent wizard	Blocked
	`/interview-prep/sessions/:id`	Analysing → configure → room → generating → ended	Blocked
	`/interview-prep/sessions/:id/report`	Report (4 tabs)	Blocked
	`/interview-prep/history`	Session list	Blocked
Attack Reasoning	`/ar/`	Domain landing (15 domains)	Blocked
	`/ar/domains/:slug`	Difficulty + start/continue + attempts	Blocked
	`/ar/attempts/:id`	Generating → diagram + MCQ + justification → evaluating → result	Blocked
	`/ar/summary`, `/ar/history`	End-of-session summary, history	Blocked
	`/ar/scenarios[/:id]`	**Orphaned legacy**	Blocked

4.4 Simulation and lab pages

App	Route / view	Purpose	Rendered
Labs hub	`/labs/`	Catalog + 4 track cards	**Yes**, 8 viewports
	`/labs/tracks/:slug`	Track listing	**Yes** (cloud, blue). An unknown slug silently shows the catalog
	`/labs/TR-CLOUD-001#/…`	AWS-style console (14 services, findings, incident center, objectives)	**Entry + layout**

App	Route / view	Purpose	Rendered
	`/labs/{soc,devsecops,aisec,threatwall,redteam}/`	Hosted labs behind the sign-in gate	Each app tested standalone, not via the hosted path
SOC	`#/`, `#/level/:id`, `#/case/:id`, `#/demo[/:id[/summary]]`, `#/badges`, `#/report`	Home, scenarios, console, guided demo, badges, final report	Home, level, demo intro, badges, case (monitor stage)
Red Team	state views: login → catalog → workspace (Browser, Proxy, Repeater, Terminal, Findings, Report)	Pentest lab	**All views**
ThreatWall	`#/start`, `#/record`, `#/notebook[/:id]`, lab (52 appliance routes)	Firewall lab	Home, record, notebook, lab entry
DevSecOps	`#/`, `#/start`, `#/labs`, `#/lab/:id`	Welcome, mode, dashboard, workspace	All four
AI Security	state views: entry → mission → briefing → workspace → results / badges / learning	LLM security lab	Entry → workspace

4.5 Reports and results

The following report and result views exist:

- InterviewPrep report and PDF
- Attack Reasoning result and summary
- TR-CLOUD-001 incident report
- SOC after-action review and final report (print and JSON)
- Red Team report tab
- AI Security results (PDF via jsPDF)
- DevSecOps completion modal (Markdown report)
- ThreatWall training record

Only the Red Team report tab and the ThreatWall record were rendered.

4.6 Settings and administration

- The admin console exists only in onboard (blocked).
- No app has a user-settings page.

- Lab-level settings (mode or unlock pickers) appear inside the labs (DevSecOps, AI Security, Red Team, ThreatWall). The AI Security "Unlock mode" select looks like an admin control, but learners can see it (UX-15).

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5. Overall UI/UX audit

5.1 Layout and visual hierarchy

ID	Sev	Finding	Impact	Evidence
UX-02	Medium	Red Team, Beginner mode, at 1366×768: the guidance banner and step list fill the viewport, so the tool tab bar starts at about y = 580 and the Proxy request/response panes sit below the fold	Learners keep scrolling between the instructions and the tool they are told to use	RC: `docs/audit-evidence/light-redteam-proxy.png`
UX-03	Medium	SOC case console below 960 px: the workflow panel moves *above* the log feed, with nested `max-height:75vh` scroll areas	On phones and tablets the logs sit under a long form, and the inner scroll areas fight the page scroll	RC: `flow-soc-case-after-4s-375x812.png`; SC `soc/client/src/styles.css:442-444`
UX-04	Medium	Attack Reasoning attempt page is one long column: prose → narrative → diagram (up to 680 px) → question	Learners scroll between the diagram and the options while writing a justification	SC `attack-reasoning/client/src/reasoning/pages.tsx:259-275`, `app.css:42`
UX-05	Medium	InterviewPrep room shows about 20 controls and badges at once (4 header badges, timer, mute/pause/end, Repeat/Skip/Ask, voice panel, auto-submit, Tips, word count, voice settings, "Type instead", transcript)	High cognitive load during a stressful timed task	SC `InterviewPrep/.../InterviewRoom.tsx:360-690`
UX-06	Low	InterviewPrep hero `h1` stays 44 px on phones, because `.hero-copy h1` outranks the ≤760 px rule	"Welcome back, {name}" wraps onto many lines at 375 px	SC `InterviewPrep/apps/web/src/ui/theme.css:140,179`
VIS-01	**High**	Attack Reasoning dimension bars use `.tr-progress > span`, but `platform.css` styles only `.tr-bar i`. The spans have no height or fill	The score breakdown and history lose their only visual comparison	SC `attack-reasoning/client/src/reasoning/pages.tsx:309,426`; `platform.css:228-232`
VIS-02	Medium	Attack Reasoning `.ar-detail` is defined twice: a 2-column grid for scenarios (`app.css:27`) and the diagram detail panel (`app.css:125`)	Above 900 px the diagram detail text may split into two columns	SC cascade / SI visual
UX-08	Low	Duplicate branding: the strip shows "ThreatReady", and right under it the console header shows "TR ThreatReady" (TR-CLOUD-001). ThreatWall's BrandRow does the same	Looks unfinished; wastes vertical space on laptops	RC `labs-console-1440x900.png`; SC `threatready-lab-001/client/src/App.tsx:123`, `ThreatWall/src/brand/Brand.tsx:9-18`

States (loading, empty, error, success, disabled):

- **Good:**
  - The Labs catalog, SOC home and level pages have loading, empty and error-with-retry states.
  - onboard Profile shows skeleton cards.
  - SOC autosaves stage drafts every 700 ms.
- **Gaps:**
  - onboard: when `/api/overview` fails, Profile says "Refresh to try again" but renders no Refresh button (FUNC-08), and the Home "Continue" strip silently disappears.
  - Admin refetches show no loading indicator.
  - AI Security shows a blank screen until `/labs/api/me` resolves, and that request has no timeout.
  - Attack Reasoning "Load more" has no loading or error state.
  - InterviewPrep report polling shows a skeleton with no message for up to 60 s.
  - Disabled buttons rarely say why: Analyse in InterviewPrep, OAuth buttons in onboard (reason only in `title`), Turnstile-gated buttons (FUNC-06).

5.2 Content and readability

ID	Sev	Finding	Evidence
CONT-01	**High**	Placeholders on legal pages: "[Registered Office Address]" (`Landing_Page-main/privacy.html:271`, `terms.html:258`) and "[To be appointed]" for the grievance officer (`privacy.html:274`). Also "Final entitlements are being confirmed" (`pricing.html:134`)	SC
CONT-02	Medium	Raw error codes reach learners: "Failed · EvidenceRequired" or "NotSupportedByEvidence" in TR-CLOUD-001 toasts (`threatready-lab-001/client/src/App.tsx:156`)	SC
CONT-03	Medium	The product status disagrees between apps. The site says Lab Simulations is "in development" in 41 places (e.g. `Landing_Page-main/index.html:54,91`), while onboard shows it "Live" and the README lists six live labs. Showcase tabs use a further pair of names ("Reason / Simulate")	SC
CONT-04	Medium	Unexplained terms and flags: "Flagged for review", "Confidence: low" and "Simplified scoring" in AR (no reviewer queue exists, per `docs/ATTACK_REASONING.md`); "Verify · X" and "Probe · X" plus Aware/Working/Proficient/Expert in InterviewPrep; IOC, Sysmon and proctree in SOC; OWASP IDs in AI Security mission control; DPI-SSL and IKEv2 in ThreatWall	SC
CONT-05	Low	SOC says "Practice records reset daily at 00:00 UTC" (`soc/client/src/Console.jsx:200`, `LevelPage.jsx:154`). Hosted mode sets `DAILY_WIPE=false` (`threatready-lab-001/server/hosted.ts:152`), so the message is false there	SC
CONT-06	Low	Confusing labels: • SOC "Delete" actually means abandon with a score of 0 (`Console.jsx:103`). • InterviewPrep "Skip" skips Alex's speech, not the question (`InterviewRoom.tsx:458-460`). • InterviewPrep "Text-only mode" appears when only 3D fails. • InterviewPrep state `completed` reads "Scoring" (`lib/format.ts:29-30`). • DevSecOps chip "Step 1 of 1" (`GetStarted.tsx:41`).	SC

ID	Sev	Finding	Evidence
CONT-07	Low	SOC page title still says "| Aerovant Technologies" (`soc/client/index.html:7`). The AI Security README calls the assistant "Nova"; the status doc says "Aero"	SC
CONT-08	Low	InterviewPrep tab-guard copy "You're not allowed to switch tabs" is punitive. It has no real consequence, and locking a phone also triggers it (`InterviewRoom.tsx:112`)	SC

5.3 Visual consistency and design system

The design system as it exists today:

- `Landing_Page-main/assets/css/site.css:8-64` holds the source tokens: navy surfaces, the `#6D5DFC` accent, `--accent-text #A59BFF` for text, Inter and JetBrains Mono, a spacing scale `--s1..--s9`, and radii `--r1/2/3`.
- `platform.css` is a shared shell copied into four apps (and drifted in Red Team), scoped to `.trp`.
- Each lab otherwise keeps its own stylesheet.

ID	Sev	Finding	Evidence
VIS-03	Medium	**Three button systems.** site `.btn` 44 px; platform `.tr-btn` 42 px; onboard `.btn` 46 px; InterviewPrep `.btn` 38 px. Each lab has its own buttons too	SC `site.css:103`, `platform.css:116`, onboard `app.css:139`, InterviewPrep `theme.css:199-221`
VIS-04	Medium	Container width is 1400 px on the site and 1280 px on the platform. Body text is 17/1.65 on the site, 16/1.6 on the platform and 1rem/1.5 in onboard. The focus ring is 2 px vs 3 px	SC
VIS-05	Medium	**Semantic colour misuse in InterviewPrep.** Errors, danger buttons, the low-time timer, scores below 50 and "Failed" all use the brand violet, while `--bad #e5665f` is defined and unused	SC `theme.css:215-216,236,326-327`, `room.css:76-77`, `components.tsx:106`, `platform.css:32`
VIS-06	Medium	`sample-report.html` uses 6 tokens that are never defined (`--primary` ×20, `--bg-subtle` ×13, `--r-sm` ×11, `--border-2` ×7, `--r-lg` ×3, `--shadow-md`) and 82 hex literals	SC; RC: 35 contrast failures on that page
VIS-07	Medium	**Theme support is fragmented.** • The site saves the theme to localStorage; onboard reads a `tr-theme` cookie. A choice on threatready.io never reaches the signed-in apps. • SOC, ThreatWall, DevSecOps and AI Security have no light theme and ignore the platform cookie. • Red Team supports light, but its raw HTTP panels and severity badges keep dark-only colours. axe found **53 contrast failures** in Red Team light mode.	SC; RC `light-redteam-proxy.png`
VIS-08	Low	ThreatWall uses IBM Plex and teal `#2dd4bf` instead of Inter and the Labs marker `#3fb6a8`. SOC keeps teal `rgba(95,201,196,…)` left over from an earlier palette, plus a pink tour accent `#ff5aa8`	SC

ID	Sev	Finding	Evidence
VIS-09	Low	Raw colour literals bypass tokens: ThreatWall 320 hex values, Labs client 227, the marketing site 222 (+193 inline `style=""`), InterviewPrep 174 (+148 inline styles). onboard `app.css` has 0 hex values (good)	Pattern count
VIS-10	Low	The Labs product marker (teal) shows only as a dot. Everything else in the catalog is accent violet, so Labs and Attack Reasoning look alike	SC `threatready-lab-001/client/src/catalog.tsx:69,107`

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6. Overflow and layout defects

6.1 Method

On every page, at every viewport, the sweep compared `document.documentElement.scrollWidth` with `clientWidth`. It then listed each element whose right edge passed the viewport and that was **not** inside a horizontally scrolling or clipping ancestor. Intentional inner scrolling (log tables, `.dt-scroll`, `.table-wrap`, terminals) was therefore not counted as page overflow.

6.2 Confirmed defects (runtime)

OVF-01: The lab-nav strip breaks full-height lab layouts · **Critical** · RC

Field	Detail
Routes	`/labs/TR-CLOUD-001` (real); ThreatWall lab, DevSecOps workspace, AI Security workspace (simulated by injecting `lab-nav.js`); SOC and Red Team consoles (SI, same CSS chain)
Viewports	1366×768 and 768×1024 (measured); all heights affected
Component and source	`threatready-lab-001/client/public/lab-nav.js:9,20` adds a 40 px sticky strip with `z-index:2147483000` and forces `html.tr-has-labnav, body{height:auto!important}`. The labs depend on `height:100%` / `100vh`: `threatready-lab-001/client/src/styles.css:3,22`, `ThreatWall/src/training/training.css:2`, `DevSecOps-Simulation/src/styles.css:128`, `soc/client/src/styles.css:32,160`, `red_team_lab/web/client/src/styles.css:5,11`
Steps	1. Open `/labs/TR-CLOUD-001` signed in. 2. Close the incident briefing. 3. Measure `.app` and `.dock`.
Expected	The console fills the viewport below the strip. `.main`, the side panel and the left nav scroll independently. The dock stays pinned at the bottom.
Actual	At 1366×768 the document is **1,528 px** tall. `.app` is 1,488 px, `.side` is 1,420 px with `overflow-y: visible`, and the **dock starts at y = 1,318**, so the page scrolls as a whole and the dock is off-screen. At 768×1024 the document is 1,703 px. With the strip injected, ThreatWall and DevSecOps measure **808 px in a 768 px viewport** (40 px too tall), and AI Security measures 964 px vs 924 px.
Second effect	With the incident-briefing modal open, `elementFromPoint` at the strip's position returns the strip. "← Back" and "All labs" stay clickable above the modal backdrop.
Evidence	`labs-console-1440x900.png` (strip undimmed above the modal); DOM measurements in §16
Root cause	The strip's global `height:auto` override cancels the percentage-height chain that every console layout relies on. Its huge z-index also sits above every modal (`.overlay` is z 50).
Impact	The core lab controls (dock, side panel, terminal input, ThreatWall URL bar and Lab Tools) move off-screen or scroll away. Leaving the lab in the middle of a modal is possible. This affects every hosted lab.

Field	Detail
Recommended fix	Have the strip publish its height as a CSS variable (`--tr-labnav-h: 40px`) and **not** override `html, body` height. Labs size their root with `height: calc(100dvh - var(--tr-labnav-h, 0px))` . Lower the strip's z-index below the modal layer, or make modal backdrops cover it. Do **not** use `overflow:hidden` on `body` as a shortcut.
Retest	At 1366×768 and 1600×900, with the strip present: `documentElement.scrollHeight === innerHeight` on each console; the dock, terminal input and URL bar are visible; with a modal open, the strip cannot be clicked.

OVF-02: ThreatWall home header buttons overflow · High · RC

Field	Detail
Routes	`ThreatWall #/start` (and `/` on first visit)
Viewports	320, 375 and 430 px wide (page is 512 px wide)
Component	`.tr-brandrow .right` (`ThreatWall/src/brand/Brand.tsx:15` , `brand.css:23,27`): a non-wrapping flex row holding the 40 px logo, the wordmark and the "Learning notebook" and "Training record" buttons
Steps	Open `#/start` at 375×812
Expected	Buttons wrap below the brand or collapse into a menu
Actual	The page scrolls sideways; "Training record" ends at x = 512 px
Evidence	`tw-home-375x812.png`
Root cause	Fixed-height single-row flex with `margin-left:auto` , no `flex-wrap` and no small-screen rule
Fix	Allow `flex-wrap: wrap` and a full-width second row below 640 px, or show icon buttons with `aria-label`
Retest	No horizontal scroll at 320, 375 or 430 px; both links visible and at least 24×24 px

OVF-03: ThreatWall training record header overflows · High · RC

Route `#/record` , 320 to 430 px wide; the page is **588 px** wide. The cause is the same `.right` row, here holding three buttons ("Home", "Learning notebook", "Back to lab") inside `span.tr-noprint` . Fix and retest as for OVF-02. Evidence: `tw-record-375x812.png` .

OVF-04: DevSecOps workspace bar pushes actions off-screen · High · RC

Field	Detail
Route	`DevSecOps #/lab/red-build` (any lab)

Field	Detail
Viewports	320 and 375 px (bar is about 406 px wide)
Component	`header.wsbar` (`DevSecOps-Simulation/src/pages/LabPage.tsx:176`, `styles.css:129,261`): "← Labs", level, mode, objectives count, timer, "Commit & Push", reset
Expected	The bar wraps, or secondary actions move into a menu
Actual	`wsbar`, `.wsmain` and `aside.task` all reach x = 406. "Commit & Push" is cut off at the right edge, and the page scrolls sideways
Evidence	`flow-devsecops-lab0-375x812.png`
Root cause	Non-wrapping flex with several `nowrap` children; the mobile sticky bar has no overflow strategy
Fix	Wrap into two rows below 480 px, or move Commit & Push and Reset into an overflow menu. Keep Commit & Push reachable
Retest	At 320 and 375 px every bar action is visible or in a reachable menu; `scrollWidth ≤ clientWidth`

OVF-05: DevSecOps mode switcher overflows the dashboard · Medium · RC

Route `#/labs`, 320 and 375 px wide; the page is 407 px wide. The cause is `div.segmented` (`DevSecOps-Simulation/src/pages/Dashboard.tsx:24`, `styles.css:123`): an `inline-flex` row with `overflow:hidden`, sitting beside a "Mode" label in a row that does not wrap. Fix: let the header row wrap so the segmented control drops onto its own row, or use a `<select>` on small screens. Evidence: `dso-dashboard-375x812.png`.

OVF-06: DevSecOps welcome chip and heading overflow at 320 px · Low · RC

Route `#/`, 320 px wide; the page is 328 px wide. The cause is the `span.chip.green` "Hands-on · no theory quizzes · runs in your browser" (`DevSecOps-Simulation/src/pages/Welcome.tsx:45`), which does not wrap, and the `h1`. Fix: allow the chip to wrap (`white-space: normal`) and cap the `h1` size at small widths. Evidence: `dso-welcome-320x568.png`.

OVF-07: SOC demo intro columns overflow at 320 px · Medium · RC

Route `#/demo`, 320 px wide; the page is 336 px wide. The cause is `.demo-cols { grid-template-columns: repeat(auto-fit, minmax(320px, 1fr)) }` (`soc/client/src/tour/tour.css:24`). With page padding, a 320 px minimum track cannot fit. The same pattern exists in `.scenarios minmax(340px)` (`styles.css:140`), `.learn-grid 300px` (`tour.css:13`) and Red Team `.cat-grid minmax(300px)` (`red_team_lab/web/client/src/styles.css:120`). Fix: `minmax(min(100%, 320px), 1fr)`. Evidence: `soc-demo-320x568.png`.

OVF-08: TR-CLOUD-001 console top bar overflows at 320 px · Medium · RC

Route `/labs/TR-CLOUD-001`, 320 px wide; the page is 356 px wide. The cause is `div.top-r` (`threatready-lab-001/client/src/App.tsx:125`, `styles.css:27`), which holds the bell, help and account name. At 375 px the account name wraps onto two lines and the service search shrinks to an empty pill. The side panel (350 px, `App.tsx:224`) and `.ddmenu.wide{width:380px}` (`styles.css:29`) are also wider than a phone (SI). Fix: collapse the top bar into a menu below 600 px, cap the panel width at `min(350px, 90vw)`, and default the nav and side panel to closed below 1100 px. Evidence: `labs-console-375x812.png`, `labs-console-320x568.png`.

OVF-09: Pilot page list item overflows at 320 px · Low · RC

Route `/pilot.html`, 320 px wide; the page is 341 px wide. An `li` containing a long unbroken email address (`Landing_Page-main/pilot.html:201`) does not wrap. Fix: add `overflow-wrap: anywhere` to body copy (site-wide, in `site.css`). Note: `site.css:70` sets `body{overflow-x:hidden}`, which clips this text instead of letting it scroll; that hides defects rather than fixing them.

OVF-10: Sample report document cards overflow on phones · Medium · RC

Route `/sample-report.html`, 320 and 375 px wide; the page is 378 px wide. The cause is `.rdoc` cards (`sample-report.html:198`, used at `:418,423`): a flex row of icon + text, with a filename that does not wrap. Fix: add `min-width:0` on the text column and `overflow-wrap:anywhere` on the filename. Evidence: `site-sample-report-375x812.png`.

OVF-11: AI Security mission control unlock select overflows at 375 px · Medium · RC

Route: mission control (state view) at 375×812; the page is 417 px wide. The cause is `label.row.small` holding a "Unlock mode:" `select` whose longest option is "Per-task (Foundation → Intermediate → Expert)" (`AI-Security/src/screens/MissionControl.tsx:70-81`). Also, the scenario table sets `minWidth: 760` (`:86`), which is intentional inner scrolling. Fix: let the select go full width on its own line below 480 px and shorten the option labels. Evidence: `flow-aisec-mission-control-375x812.png`.

6.3 Confirmed clipping (content hidden, no page scroll)

ID	Sev	Finding	Evidence
OVF-12	**High**	The DevSecOps code editor clips long lines. The textarea has `white-space:pre; overflow-x:hidden`, with `scrollWidth` 476 and `clientWidth` 354 at 375 px, so **122 px of code is unreachable**	RC; `DevSecOps-Simulation/src/styles.css:177`, `Editor.tsx:58`; `flow-devsecops-lab0-375x812.png`
OVF-13	Medium	The Red Team terminal's xterm screen (384 px) is wider than its host (341 px) at 375 px, so the right edge of each output line is cut off	RC; `red_team_lab/web/client/src/tools.tsx:220-233`; `flow-redteam-tab-terminal-375x812.png`
OVF-14	Medium	ThreatWall's simulated URL-bar placeholder is truncated on phones, and the Lab Tools tooltip overlaps the page	RC; `flow-threatwall-lab-375x812.png`

6.4 Source-inferred overflow risks (verify before fixing)

ID	Sev	Risk	Evidence
OVF-15	High	InterviewPrep has two sticky bars totalling 116 px (header 64 + sub-nav 52). The configure summary (`top:24px`) and the answer column (`top:20px`) will slide under them on desktop	SI `platform.css:141`, `theme.css:101,406`, `room.css:46`
OVF-16	Medium	onboard admin range picker: `.seg` sets `overflow-x:auto`, then `overflow:hidden` in the same rule, so on phones the last option ("12 months") is likely clipped and cannot be scrolled to. This contradicts prior-audit M3 "Fixed"	SC `apps/web/src/styles/app.css:1379,1384`
OVF-17	Medium	InterviewPrep has no `overflow-wrap` / `word-break` anywhere. Long URLs or stack traces in transcript bubbles or report answers may overflow	SC
OVF-18	Medium	Red Team browser bar `.bchrome` is a non-wrapping row (method select, host, URL, Go, Intercept) where `.grow` lacks `min-width:0`. It may overflow below about 420 px	SI `styles.css:112,200`
OVF-19	Medium	AI Security inline `gridTemplateColumns:"1fr 1fr"` in Results and Learning cannot be overridden by media queries, so those screens stay two-column on phones	SC `Results.tsx:75`, `Learning.tsx:140`
OVF-20	Low	ThreatWall toasts use `right: 372px`, which is not adjusted for narrow layouts, so on phones they may render off-screen	SC `ThreatWall/src/index.css:210`
OVF-21	Low	Attack Reasoning diagram: edge labels are ellipsised with `pointer-events:none` and no `title`, so truncated labels can't be revealed; wide graphs at `minZoom 0.2` may be unreadable on phones	SC `app.css:137`, `Diagram.tsx:282-301`
OVF-22	Low	Attack Reasoning difficulty selector `.ar-levels` is `inline-flex; overflow:hidden`. Three buttons measure about 299 px, which may clip "Advanced" at 320 px	SI `app.css:26,130`

7. Usability and cognitive load

7.1 Orientation questions

The table answers six questions a learner should be able to answer at any point:

- **Where?** Do I know where I am?
- **Next?** Do I know what to do next?
- **Status?** Can I see my progress or state?
- **Success?** Can I tell an action worked?
- **Recovery?** Can I recover from an error?
- **Back?** Can I go back safely?

Key: ✓ = yes · ◐ = partly · ✗ = no.

App	Where?	Next?	Status?	Success?	Recovery?	Back?
Marketing site	✓ breadcrumbs	✗ CTAs broken	n/a	n/a	n/a	✓
onboard	◐ "Home" no longer marked current everywhere (fixed)	◐ Continue strip vanishes on error	✓	✓	◐ no retry on Profile	✓
InterviewPrep	◐ no `h1` or page title on several screens	✓ stepper	◐ timer low state is colour only	✓ toasts	◐ the "Try again" error action only clears the error	✗ wizard lost on reload
Labs catalog	✓	✗ Blue Team looks empty	◐ 5 of 6 labs show no progress	—	✓	✓
TR-CLOUD-001	✓ simbar + level	✓ objectives	✓	◐ raw error codes	✓	◐ "archived" runs can't be opened
Attack Reasoning	✓	✓	✗ bars invisible (VIS-01)	◐ result not announced	✓ Retry	✗ Skip loses the justification
SOC	✓ stage rail	✓	✓	✓ field verdicts	◐ "Reconnecting…" with no limit	✓
Red Team	◐	✓ Beginner / ◐ Advanced	◐ objective state shown by colour only	◐ 4.2 s toasts	◐	✗ reload returns to catalog

App	Where?	Next?	Status?	Success?	Recovery?	Back?
ThreatWall	✓	✓ "Next:"	✓ verification list	✓	✓ Re-verify	✓
DevSecOps	◐ no `h1`	◐ no "why not yet"	✓	◐ no live region	◐ `window.confirm`	◐
AI Security	✓	✓	✓	✓	✗ cannot finish a failed scenario	✗ Back leaves the app

7.2 Findings

ID	Sev	Finding	User impact	Recommendation	Evidence
UX-01	**High**	The Blue Team track card says "Coming soon", and `/labs/tracks/blue-team` says "No labs in this track yet". The SOC lab (track `blue`) is live but flagged `home`, and home labs are excluded from tracks. The Cloud Security blurb promises DevSecOps for the same reason	Learners conclude there is no Blue Team content	List `home` labs in their track as well	**RC** `labs-track-blue-1440x900.png`; SC `catalog.tsx:80-86,99,116`, `server/catalog.ts:22`
UX-07	**High**	InterviewPrep voice auto-submit is on by default: 5.5 s of silence plus a 4 s countdown, with no live region. Its "Keep talking" button actually turns auto-submit off for the rest of the session	Slow speakers, people who stutter and screen-reader users lose answers	Make it opt-in or adjustable (≥20 s); announce the countdown; label the button truthfully	SC `InterviewRoom.tsx:49-50,144,285-301,547`
UX-09	High	Attack Reasoning "Skip to next scenario" discards a typed justification (up to 4,000 characters) with no confirmation, and the draft is not saved	Lost work	Confirm when text exists; keep the draft in sessionStorage per attempt	SC `pages.tsx:214-218,274`
UX-10	Medium	Red Team reload always shows the catalog again. Picking a different mode then silently resumes the old run in the old mode, while the button says "Start engagement · Advanced"	Disorienting; the UI misstates what will happen	Resume the active run directly; never show a mode that won't be applied	SC `App.tsx:59,79,121`; `server/app.ts:130`
UX-11	Medium	TR-CLOUD-001: viewing a resource advances the simulated clock and the attacker, but the copy says the clock moves "when you act"	Feels unfair; learners are punished for reading	Make inspection free, or say so plainly	SC `server/engine.ts:536-538`; `App.tsx:126`
UX-12	Medium	InterviewPrep loses the wizard, configure choices and the report tab on reload	Rework	Put the step and document ids in the URL; report tab in `?tab=`	SC `CreateInterviewPage.tsx:12-18`, `ReportPage.tsx:54`
UX-13	Medium	Red Team reveals each finding's true severity (badge + prefilled select) before the learner documents it	Removes a judgement the lab is meant to assess	Hide it until submitted	SC `tools.tsx:285,290`
UX-14	Medium	DevSecOps lets learners switch mode mid-lab with no warning (Guided hints, then Expert bonus). The Expert timer keeps counting while the tab is closed	Unclear rules; score feels arbitrary	Lock the mode per attempt or confirm a switch; pause when hidden	SC `Dashboard.tsx:24-30`, `engine/lab.ts:158-160`, `LabPage.tsx:188`
UX-15	Low	AI Security "Unlock mode" is a select learners can see; it reads like an admin setting	Confusion	Move it into a settings disclosure with an explanation	RC/SC `MissionControl.tsx:70-81`

ID	Sev	Finding	User impact	Recommendation	Evidence
UX-16	Medium	Feedback on important outcomes disappears quickly: • Red Team: "Re-test FAILED" and wrong-fix messages vanish in 4.2 s toasts. • InterviewPrep: error toasts last 6 s (including "Microphone blocked").	Missed information; WCAG 2.2.1	Keep a persistent result line per item; errors don't auto-dismiss	SC `red_team_lab/.../ui.tsx:10-14`; `InterviewPrep/.../components.tsx:236`
UX-17	Low	Unknown routes fall back silently: SOC `#/level/<bad>` shows home, DevSecOps `#/lab/<bad>` shows Welcome, Labs `/tracks/<bad>` shows the catalog. The URL stays wrong	Broken bookmarks give no signal	Show a not-found message with a link	**RC** (all three)
UX-18	Medium	Mobile users can't reach key actions: • Red Team "Reset" and "Brief" are hidden at ≤720 px with no alternative. • DevSecOps hides the file tree at ≤720 px, so arbitrary files can't be opened. • The DevSecOps splitter is mouse-only.	Tasks can't be completed on phones, and nothing says so	Move actions into a menu; add a file picker; or show a "best on a larger screen" notice	**RC** (Reset/Brief hidden, file tree `display:none`); SC
UX-19	Low	ThreatWall logs the learner out of the simulated appliance after 5 idle minutes, even while the learner is reading the task panel	Interrupts learning	Pause the idle timer while the task panel has focus, or warn first	SC `state/baseline.ts:104`
UX-20	Low	SOC stage rail labels read "Tier 1 · Tier 2 · Tier 1" (Monitor, L1 triage, L2 review, Closure)	May confuse learners about who closes the ticket	Confirm intent; if L1 closes, label it "Tier 1 · Closure"	RC `flow-soc-case-after-4s-1366x768.png` (product decision)

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8. Learning and simulation workflow review

Workflow	Entry	Objective	Sequence (actual)	Success state	Error / recovery	Persistence	UX concerns	Tested
Labs catalog → lab	`/labs/`	Pick a lab	catalog → track → card → TR-CLOUD-001 SPA, or a hosted lab via full-page link	Lab opens	Retry on catalog error; an unknown lab id shows "Cannot load the lab"	Opening TR-CLOUD-001 **creates a run**, so the card shows "In progress · 0%"	UX-01; only 1 of 6 labs shows progress; no cross-lab prerequisites (by design)	RC catalog/tracks
TR-CLOUD-001	`/labs/TR-CLOUD-001`	Investigate, contain, secure	2-step briefing → objectives (locked by level, server auto-advances) → inspect evidence → report finding (modal) → evidence locker → incident report (80+ chars, confirm)	Level-up modal; after completion only "Open the incident report from your profile menu"	Raw error codes (CONT-02); reset by typing RESET; "archived" runs can't be viewed	Server-side with idempotency keys; editor drafts in memory only (guarded by `beforeunload`)	OVF-01, OVF-08, UX-11; change buttons stay enabled after completion	RC briefing + layout
Attack Reasoning	`/ar/domains/:slug`	One AI-generated assessment	difficulty → Start → generating (polls; after 75 s "you can leave") → diagram + MCQ + justification → submit → evaluating (3 stages) → result → Next/End → summary	Result view; summary	Retry on `generation_failed` / `evaluation_failed`; no End button on `generation_failed`	Attempt server-side; selection and text client-only	VIS-01, UX-09, A11Y-08 (states not announced), duplicate polling	Blocked
SOC case	level → Start	Triage → investigate → respond	Monitor queue → acknowledge → L1 form (autosave) → L2 review rounds → L3 response (server-confirmed risky actions) → after-action review	Result banner (`role=status`) + review	409 "attempt open" → resume or abandon; stream reconnect loops with no limit and no backfill	Server-side; drafts autosaved	FUNC-02 (hosted), UX-03, stream gaps not shown	RC through monitor stage; stream "Live" standalone
SOC guided demo	`#/demo`	Learn the SOC flow	16-step instructor overlay	Demo summary	Pause, exit, resume	Server-side	OVF-07	RC intro
Red Team TR-RED-001	catalog → Start engagement	Recon → exploit → remediate → report	Briefing (must acknowledge) → Browser / Proxy / Repeater / Terminal → findings (verify fix) → one-shot report	"Report quality score x/y"	Wrong fix → 4.2 s toast; report can't be resubmitted and has no confirmation	Run server-side; terminal, Repeater edits and report draft client-only	UX-02, UX-10, UX-13, A11Y-01/02, OVF-13; score breakdown not shown	RC all tabs

Workflow	Entry	Objective	Sequence (actual)	Success state	Error / recovery	Persistence	UX concerns	Tested
ThreatWall	`#/start` → Start lab	Configure a firewall, levels 0–4	Ordered tasks; steps auto-verify from config state; "Show me", Hint (Practice), Got it	"Task complete" + Next; Re-verify if something later breaks	Exact failure reasons (`role=alert`)	localStorage (**not per user**, FUNC-09); a version bump silently discards progress	OVF-02/03, UX-19; strongest feedback model in the platform	RC home → lab
DevSecOps	Welcome → mode → dashboard → lab	Fix pipelines and vulnerabilities	Objectives (boolean) + editor + terminal + pipeline; Commit & Push modal; completion modal with Markdown report	Completion modal	`window.confirm` resets; no "why not yet"	localStorage (**not per user**); opening a locked lab URL writes state	OVF-04/05/12, UX-14, A11Y-01; learner code runs on the main thread (a loop can freeze the tab, SI)	RC through lab 0
AI Security	Entry → mission control → Start	Attack, detect, fix, re-test	Briefing → workspace tabs (target, map, logs, guardrails, detection, re-test, report) → Finish	Results with breakdown and "Why you got this score"	**Failed re-test: no finish path** (FUNC-04)	Progress per platform user; **active attempt in memory only** (FUNC-05)	A11Y-07 (map), OVF-11/19	RC to workspace + reload
InterviewPrep	dashboard → New	Practise an interview	resume → JD + consent → analysing (SSE) → configure → room (voice/typing, timer) → evaluating → report → PDF	Report	Mic `audio-capture` / `other` errors show nothing; PDF button only after a reload	Sessions server-side; wizard client-only	UX-05/07/12, VIS-05, FUNC-07	Blocked
onboard sign-up/sign-in	`/`	Get an account	username + email + password + Turnstile → verify OTP → sign in again	`/app`	Turnstile failure leaves buttons disabled with no message (FUNC-06)	—	Retyping the password after verify; social sign-in below the form (prior L1)	Blocked
Trial	`/` → trial	Try without email	Trial session → products say "Needs a free account"	—	—	Anonymous user	FUNC-03	Blocked

9. Responsive testing

9.1 Route × viewport matrix (horizontal-overflow criterion)

Statuses are defined in §2.4. **FAIL** shows the measured page width. "NOT TESTED" marks combinations that were not run.

Page / state	URL	320×568	375×812	430×932	768×1024	1024×768	1366×768	1440×900	1920×1080
Marketing: 15 pages (index, attack-reasoning-lab, lab-simulations, interviewprep, pricing, transparent-scoring, rubric, how-it-works, resources, contact, security, privacy, terms, dpa, 404)	`/*.html`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Marketing: pilot	`/pilot.html`	**FAIL** (341)	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Marketing: sample report	`/sample-report.html`	**FAIL** (378)	**FAIL** (378)	PASS	PASS	PASS	PASS	PASS	PASS
Labs catalog	`/labs/`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Labs track: Cloud Security	`/labs/tracks/cloud-security`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Labs track: Blue Team	`/labs/tracks/blue-team`	PASS*	PASS*	PASS*	PASS*	PASS*	PASS*	PASS*	PASS*
TR-CLOUD-001 console (briefing open)	`/labs/TR-CLOUD-001`	**FAIL** (356)	PASS	PASS	PASS†	PASS†	PASS†	PASS†	PASS†
ThreatWall home	`#/start`	**FAIL** (512)	**FAIL** (512)	**FAIL** (512)	PASS	PASS	PASS	PASS	PASS
ThreatWall record	`#/record`	**FAIL** (588)	**FAIL** (588)	**FAIL** (588)	PASS	PASS	PASS	PASS	PASS
ThreatWall notebook	`#/notebook`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
ThreatWall lab	`/` (after Start)	NOT TESTED	PASS	NOT TESTED	PASS	NOT TESTED	PASS†	NOT TESTED	NOT TESTED
DevSecOps welcome	`#/`	**FAIL** (328)	PASS	PASS	PASS	PASS	PASS	PASS	PASS

Page / state	URL	320×568	375×812	430×932	768×1024	1024×768	1366×768	1440×900	1920×1080
DevSecOps mode choice	`#/start`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
DevSecOps dashboard	`#/labs`	**FAIL** (407)	**FAIL** (407)	PASS	PASS	PASS	PASS	PASS	PASS
DevSecOps workspace	`#/lab/red-build`	**FAIL** (407)	**FAIL** (406)	PASS	PASS	PASS	PASS†	PASS	PASS
AI Security entry	`/`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
AI Security mission control	(state)	NOT TESTED	**FAIL** (417)	NOT TESTED	PASS	NOT TESTED	PASS	NOT TESTED	NOT TESTED
AI Security briefing / workspace	(state)	NOT TESTED	PASS	NOT TESTED	PASS	NOT TESTED	PASS	NOT TESTED	NOT TESTED
SOC home / level / badges	`#/`, level, `#/badges`	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
SOC demo intro	`#/demo`	**FAIL** (336)	PASS	PASS	PASS	PASS	PASS	PASS	PASS
SOC case console	`#/case/:id`	NOT TESTED	PASS‡	NOT TESTED	PASS‡	NOT TESTED	PASS	NOT TESTED	NOT TESTED
Red Team login / catalog / 6 tabs	`/lab/`	PASS (login)	PASS	PASS (login)	PASS	PASS (login)	PASS	PASS (login)	PASS (login)
onboard (all routes)	—	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED
InterviewPrep (all routes)	—	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED
Attack Reasoning (all routes)	—	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED	BLOCKED

* Passes the layout check but shows the wrong content (UX-01). † Passes standalone. **Fails vertically once the hosted lab-nav strip is present** (OVF-01). ‡ No sideways scroll, but usability is poor: workflow above logs, nested scrolling (UX-03).

9.2 Other responsive observations

- **Touch targets under 24×24 px at 375 px:**
  - Marketing legal pages: 23–38 per page (mostly inline links in body text, which WCAG 2.5.8 exempts).
  - ThreatWall notebook: 40.
  - The SOC and Labs consoles: 3–10.

- **Sidebar collapse:**
  - The Labs console nav and side panel become overlays below 1100 and 760 px but **default to open** (SI).
  - Red Team rails become off-canvas drawers below 1100 px, but those drawers stay focusable (A11Y-03).
  - ThreatWall's sidebar becomes a drawer below 860 px.
- **Desktop-only apps don't say so:**
  - ThreatWall's only hint is "Best viewed at 1920×1080" on its simulated login screen.
  - DevSecOps, Red Team and SOC show no notice at all.
- **Not tested:** orientation changes, browser zoom, virtual keyboard.

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10. Accessibility

10.1 Automated scan (axe-core 4.10.3, 36 pages)

Rule	Impact	Pages	Nodes	Main sources
`color-contrast`	serious	16	**241**	violet `#6D5DFC` as text (SOC, DevSecOps, AI Security, rubric); DevSecOps gutter `#3f4b69` (2.2:1); sample report greens (3.57:1); ThreatWall `#6f8597` meta text (4.36:1); console logo white on `#ec7211` (3.0:1)
`region`	moderate	15	165	content outside landmarks (no `<main>` in SOC pages, DevSecOps, ThreatWall home/record)
`link-in-text-block`	serious	11	18	site breadcrumb links distinguishable only by colour (1.53:1 vs surrounding text)
`landmark-one-main`	moderate	14	18	missing `<main>`
`landmark-no-duplicate-banner` / `landmark-unique`	moderate	9 / 6	16	site legal pages; ThreatWall `role="banner"` on BrandRow
`heading-order`	moderate	7	7	h4 after h2 on legal and notebook pages
`page-has-heading-one`	moderate	3	4	DevSecOps mode choice and workspace, Red Team
`label`	**critical**	1	1	DevSecOps editor textarea
`link-name`	serious	1	1	—

The automated scan could not run on onboard, InterviewPrep or Attack Reasoning (blocked).

10.2 Findings

ID	Sev	WCAG	Finding	Evidence
A11Y-01	**High**	2.1.2	**Keyboard traps.** • DevSecOps editor: Tab always calls `preventDefault`; after pressing Tab, focus stays in `TEXTAREA`. • Red Team terminal: after two Tabs, focus stays in `xterm-helper-textarea`. • DevSecOps terminal: same handler.	**RC**; `DevSecOps-Simulation/src/components/Editor.tsx:60-69`, `Terminal.tsx:111-113`; `red_team_lab/.../tools.tsx:244`

ID	Sev	WCAG	Finding	Evidence
A11Y-02	**High**	1.3.1, 4.1.2, 2.1.1, 4.1.3	**Red Team has no accessibility baseline.** • One ARIA attribute in the whole client. • Inputs with no label: login, mode select, URL, method, cookies, `Field` labels without `htmlFor`. • Mouse-only controls: link chips are `<span onClick>`, proxy rows are `<tr onClick>`, guide links are `<a>` without `href`. • Modals with no dialog role, focus handling or Escape. • Toasts with no live region. • "✕" buttons with no name.	SC `App.tsx:49,181,226,366-368,394`, `tools.tsx:101-139,132,157`, `ui.tsx:17,52`
A11Y-03	High	2.4.3, 2.4.7	Red Team's closed drawers are moved off-screen (left −322 px / right 375–698 px) but remain `visibility:visible`, not `inert` and not `aria-hidden`. They contain 3 and 2 focusable controls, so keyboard focus can land on invisible controls	**RC** at 375 px
A11Y-04	High	2.4.3	Modals across the platform lack a focus trap: Labs `ui.tsx`, SOC `BadgeDialog`, ThreatWall `Dialog`, InterviewPrep tab guard. DevSecOps and AI Security modals also lack Escape and initial focus	SC (per app; see the source reviews in §16)
A11Y-05	**High**	1.4.3	**Brand violet used as text.** `#6D5DFC` on `#0b1020` is 4.17:1 and on `#111827` is 3.90:1. Used for SOC links, tabs and "Sign out"; DevSecOps `.lvl` and `.who`; AI Security links and `summary`; onboard links, `.link-btn` and the skip link; rubric bands. The platform already defines `--accent-text #A59BFF` (7.4:1)	**RC** (axe) + SC `onboard app.css:21,78,214`
A11Y-06	Medium	1.4.11	Input borders fail 3:1 against their background: onboard 1.73:1 (dark) and 1.43:1 (light); InterviewPrep 1.85:1 and 1.51:1. AI Security's switch-off track is 2.43:1	SC (computed)
A11Y-07	High	2.1.1, 4.1.2	AI Security architecture map: nodes are `<g role="button">` with no `tabIndex` or key handler, inside `<svg role="img">`. The **scored** fix-layer step can't be done by keyboard or screen-reader users. The Labs architecture SVG has the same `role="img"` -with-buttons problem	SC `AI-Security/src/.../ArchitectureMap.tsx:80,99-105`; `threatready-lab-001/.../architecture.tsx:294,314`
A11Y-08	Medium	4.1.3	Async states are not announced: • Attack Reasoning generating, evaluating and result arrival (no `role=status`, focus not moved). • InterviewPrep auto-submit countdown, low time and interim transcript. • SOC stream status. • DevSecOps: no live regions at all. • AI Security chat log and re-test results.	SC
A11Y-09	Medium	1.3.1, 4.1.2	ARIA tab and radio patterns are incomplete: `role=tablist` with no tabpanels and no arrow keys in onboard auth, InterviewPrep, SOC and Labs; `role=radio` without roving focus in AR and InterviewPrep; `aria-selected` on `role=radio` in InterviewPrep	SC
A11Y-10	Medium	2.4.2, 2.4.3	No per-route `document.title` and no focus move on route change in InterviewPrep. Several screens lack an `h1` (InterviewPrep analysing/room, DevSecOps workspace, Red Team)	SC + RC (axe `page-has-heading-one`)

ID	Sev	WCAG	Finding	Evidence
A11Y-11	Medium	1.4.1	Site breadcrumb links are told apart from surrounding text only by colour (1.53:1). Inline links in DPA and sample report: 1.05:1	**RC** (axe, 11 pages)
A11Y-12	Low	1.3.1	Site legal pages have duplicate `banner` landmarks and skip heading levels; the mobile drawer's `h3`s come before the page `h1`	RC (axe) + SC
A11Y-13	Medium	2.4.1	onboard skip link (`z-index:10`) is likely hidden under the sticky header (z 50)	SI `app.css:99-110`, `platform.css:141`
A11Y-14	Medium	1.1.1	Diagram and chart alternatives are missing: AR canvas is `role="img"` with a generic label; onboard admin chart values live only in `title`; AR technique chips show the name only in `title`	SC
A11Y-15	Low	1.4.1	Colour-only status: Red Team objective completion (green dot); InterviewPrep low-time timer; DevSecOps file-tree M/U letters and mode selection with no `aria-pressed`	SC
A11Y-16	Low	2.3.3	`prefers-reduced-motion` is ignored by ThreatWall, DevSecOps and AI Security, and by the InterviewPrep 3D avatar idle motion. The platform shell, SOC and onboard honour it	SC
A11Y-17	Low	1.1.1	Emoji carry meaning and are read aloud: 🔒 is the only lock indicator in Red Team; 📄 ⚙️ 💡 in DevSecOps	SC

Strengths worth keeping:

- InterviewPrep always shows questions as text and keeps a full transcript, which works for deaf and hard-of-hearing users.
- SOC severity always includes text, and its log rows expand through real buttons.
- ThreatWall uses `role="status"` / `role="alert"` correctly.
- The platform shell has a skip link, labelled navs and `main#main`.

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11. Functional UI validation

ID	Sev	Control or flow	Expected	Actual	Evidence
FUNC-01	**Critical**	Marketing "Sign in" and "Start free" (about 74 links)	Open onboard sign-in or sign-up	They point to `https://threatready.io/app/…` and `/app/auth?mode=signup&plan=free`. No `/app` exists on the site, onboard has no `/app/auth` route, and `404.html` rewrites `/app/*` to `/app/?/…`, which may loop	SC `Landing_Page-main/index.html:79-80,126`, `404.html:9-16`; PA (security audit)
FUNC-02	**Critical**	SOC live stream, case export, record export (hosted)	Reach the SOC API under `/labs/soc/`	Absolute `/api/cases/:id/stream`, `/api/cases/…/export` and `/api/me/export` paths go to the onboard API (404/401); EventSource retries forever	SC `soc/client/src/CaseView.jsx:102`, `components/ReportPanel.jsx:125-126`, `pages/ReportPage.jsx:19`; PA. Standalone, the stream works ("Live", RC)
FUNC-03	High	Trial	Use the promised trial allowance	Every product says "Needs a free account"; returning to an app is refused. The site FAQ promises something different again	SC `AuthPage.tsx:792-795`, `ProductPage.tsx:302-310`, `AppShell.tsx:117-121`, `index.html:34`
FUNC-04	High	AI Security "Finish & see results" after a failed re-test	A failed attempt still produces a result (README)	The button appears only when the re-test passes, so there is no way out	SC `RetestRunner.tsx:77-87` vs `README.md:57`
FUNC-05	High	AI Security reload, Back, Exit mid-scenario	Resume the attempt	**Reload returns to Mission Control; the attempt is lost.** Views are not URLs, so Back leaves the app; Exit has no confirmation	**RC** (reload); SC `state/store.tsx:160-176`, `Shell.tsx:47-51`
FUNC-06	High	onboard forms when Turnstile fails to load	Explain and offer a retry	`useTurnstile` exposes `failed`, but nothing reads it; buttons stay disabled with no message	SC `lib/turnstile.tsx:270-273,289`; `AuthPage.tsx:277,361,541,695,799`
FUNC-07	Medium	InterviewPrep PDF button	Appears when the PDF is ready	The report is fetched once, so the button never appears without a reload	SC `ReportPage.tsx:103`
FUNC-08	Medium	onboard Profile when the overview fails	Retry control	The notice says "Refresh", but the button renders only when data exists	SC `ProfilePage.tsx:77-84`
FUNC-09	High	ThreatWall and DevSecOps progress	Per learner	localStorage keys are not namespaced by user, so the next person on the same browser continues the previous learner's lab. AI Security falls back to a shared key if `/me` fails	SC `ThreatWall/src/state/store.tsx:8`, `DevSecOps-Simulation/src/store.ts:19-20`, `AI-Security/src/engine/progress.ts:61`; PA
FUNC-10	Medium	onboard admin role change	Retype the email to confirm	The confirm field's placeholder *is* the email	SC `AdminUserPage.tsx:359`

ID	Sev	Control or flow	Expected	Actual	Evidence
FUNC-11	Medium	InterviewPrep mic errors	Every error is explained and falls back to typing	`audio-capture` and `other` show nothing; voice mode stays on. `browser-notice.js` recommends Firefox, which has no speech recognition	SC `useVoice.ts:325`, `InterviewRoom.tsx:135,243-246`, `public/browser-notice.js:17`
FUNC-12	Medium	Red Team report submit	Confirm before a one-shot action; show a breakdown	No confirmation; only the total is shown, although the server computes penalties	SC `tools.tsx:335,348`; `engine.ts:202,260-268`
FUNC-13	Low	Attack Reasoning `/scenarios` "Start scenario"	Starts something	No handler (orphaned page)	SC `App.tsx:112`
FUNC-14	Low	DevSecOps locked lab URL	Blocked with no side effects	It writes state, so the dashboard shows "In progress"	SC `LabPage.tsx:111-114` vs `145`
FUNC-15	Low	TR-CLOUD-001 change buttons after completion	Disabled	Enabled, then fail with a `LabComplete` toast	SC `services1.tsx:32`
FUNC-16	Low	Labs reset "previous run is archived"	Archived runs can be opened	No UI uses `/labs/:id/runs`	SC `App.tsx:97`, `server/app.ts:147`
FUNC-17	Low	Sign-out across labs	Consistent	Red Team "Sign out" ends the platform session; SOC hides sign-out when hosted	SC

Buttons that were checked and work in a browser:

- Red Team dev login, Start engagement, briefing acknowledge, all six tool tabs and proxy capture.
- AI Security entry, Start and Start scenario.
- ThreatWall Start lab.
- SOC Open level and Start (creates a case; stream connects).
- DevSecOps mode persistence.
- The Labs catalog loading the console and showing the briefing.

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12. Browser, console and regression testing

12.1 Console and network (runtime pages)

App	Errors observed	Assessment
Marketing site	None	Clean
Labs catalog / console	None (after API start-up)	Clean
SOC	None	Clean
DevSecOps	None	Clean
AI Security	None	Clean
ThreatWall notebook	React warning: *"Encountered two children with the same key"*	**Low (CONS-01):** duplicate keys in a notebook list can cause rows to be duplicated or dropped on update
Red Team	`401 GET /lab/api/me` before sign-in	Expected flow; an unhandled 401 shows up as a console error (cosmetic, CONS-02)

12.2 Existing automated coverage

Asset	What it checks	Gaps
`e2e/cross-browser.mjs`	Sign-in, home, theme, menus, profile, admin, InterviewPrep, labs and AR in Chromium, Firefox and WebKit; `noOverflow` at desktop and phone width	Only 2 widths; never visits AR attempt or result pages, the console at narrow widths, Blue Team track content or hosted lab layouts with the strip; no axe; no keyboard checks
`ThreatWall/e2e`	All 30 tasks	Only a 1600×900 viewport
Unit tests	InterviewPrep 38, Labs 46, AR 7, ThreatWall 46 (per docs; **not run in this audit**)	DevSecOps has none

12.3 Proposed regression suite (for after fixes)

1. Extend `e2e/cross-browser.mjs` with the **8-viewport overflow sweep** from this audit, covering both horizontal overflow and a vertical check (`scrollHeight === innerHeight`) on console screens **with the lab-nav strip present**.
2. Add an **axe-core** pass (WCAG 2.1 AA) on every route at 1440 and 375 px, failing on `critical` / `serious`.
3. Add **keyboard journeys**:

    - Tab through each lab and assert focus can leave the editor and terminal.
    - Assert no focusable element sits outside the viewport while a drawer is closed.
4. Add **workflow assertions**:
    - The Blue Team track lists SOC.
    - Reloading AI Security keeps the attempt.
    - AR Skip asks for confirmation when text exists.
    - SOC hosted stream URLs start with `BASE_URL`.
    - No marketing link contains `threatready.io/app`.
5. Run it in Chromium, Firefox and WebKit at 375×812 and 1366×768 as a minimum.

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13. Re-check of the 14 Oct 2026 UX audit

Item	Claimed	Current status	Evidence
C2 App switcher in account menu below 900 px	Fixed	**Confirmed** (source)	`platform.css:192-195`
H1 Home leads with "Continue"	Fixed	**Partly.** The strip exists but disappears on error; the hero is still marketing copy	`ProductPage.tsx:38,102-104`
H2 Product names unified	Fixed	**onboard only.** The site still says "in development" (41 places) and uses "Reason / Simulate"	CONT-03
H3–H7	Fixed	Confirmed in source	onboard review
M1 Gradients and glows removed (InterviewPrep)	Fixed	**Partly.** ScoreRing gradient + drop-shadow, glowing question dot, stage overlays and the pulsing mic halo remain	`components.tsx:116-132`, `room.css:14,33,50,72,99-100`
M3 Admin range picker scrolls	Fixed	**Not effective:** `overflow:hidden` overrides `overflow-x:auto`	OVF-16
M4 Reduced motion	Fixed	Confirmed for shell, onboard and InterviewPrep; **not** in ThreatWall, DevSecOps or AI Security	A11Y-16
M2, M5–M8	Fixed	Confirmed in source (M2 now applies only to the orphaned `/scenarios` page)	—
L1 Social sign-in first	Open	Still open	`AuthPage.tsx:194-197`
L2 InterviewPrep has no light theme	Open	**Out of date.** A light theme now exists, with gaps (dark chips, overlays, tab-guard backdrop)	VIS-07
L3 Interstitial before the AWS console	Open	**Mostly addressed** by the 2-step incident briefing	RC `labs-console-1440x900.png`
"No horizontal overflow at 375 and 768 px" (validation)	—	Holds for the pages it covered. **Ten other pages fail** (§9)	—
"Shared files identical"	—	True for 4 apps; **Red Team's** `platform.css` **has drifted**	§3.2

14. Prioritised fix plan and acceptance criteria

Rules for implementation:

- Fix the shared cause rather than each symptom.
- Never hide overflow with `overflow-x:hidden`.
- Preserve every piece of technical content.
- Retest each item at the viewports named.
- Implementation starts **only after this plan is approved**, phase by phase.

Effort: **S** ≤ ½ day · **M** ≤ 2 days · **L** > 2 days.

Phase 0: Decisions and quick wins (about 3 days)

Order	IDs	Change	Files (indicative)	Effort	Acceptance criteria
0.1	FUNC-01	Point all CTAs to onboard (`/` and `/?mode=signup`); remove the `/app` hack from `404.html` ; drop or implement `plan=`	`Landing_Page-main/*.html` , `404.html`	S	`grep "threatready.io/app"` returns 0; each CTA opens the correct onboard tab
0.2	CONT-01	Fill in or remove the legal placeholders	`privacy.html` , `terms.html` , `pricing.html`	S (needs business input)	`grep -E "\[[A-Z][^]]+\]"` returns 0
0.3	FUNC-02, CONT-05	Prefix SOC stream and export URLs with `import.meta.env.BASE_URL` ; hide the "reset daily" text when the daily wipe is off	`soc/client/src/CaseView.jsx` , `ReportPanel.jsx` , `ReportPage.jsx` , `Console.jsx`	S	Under `/labs/soc/` , the stream connects within 2 s and all downloads return 200
0.4	UX-01	Include `home` labs in their track list; fix the Cloud Security blurb	`threatready-lab-001/client/src/catalog.tsx`	S	Blue Team card shows "1 live" and the track page lists SOC
0.5	VIS-01, VIS-02	Use `.tr-bar` markup for AR bars; rename the diagram's `.ar-detail`	`attack-reasoning/client/src/reasoning/pages.tsx` , `app.css`	S	Bars are filled in proportion in both themes; the diagram detail is one column at 1280 px
0.6	A11Y-05	Replace `#6D5DFC` -as-text with `--accent-text` (or the app's equivalent) in SOC, DevSecOps, AI Security and onboard links	the respective stylesheets	S	axe `color-contrast` reports 0 nodes caused by accent text on the pages in §10.1
0.7	FUNC-03	**Product decision:** let trial sessions into the apps, or rewrite the trial copy and the FAQ	onboard `AuthPage.tsx` , site `index.html`	S–M	Trial copy matches what a trial user can actually do, everywhere

Phase 1: Layout and overflow (about 4 days)

Order	IDs	Change	Effort	Acceptance criteria
1.1	OVF-01	Lab-nav height contract: the strip sets `--tr-labnav-h` and stops overriding `html, body` height; consoles use `calc(100dvh - var(--tr-labnav-h,0px))`; strip z-index below modals	M	At 1366×768 and 1600×900 on TR-CLOUD-001, ThreatWall, DevSecOps, AI Security, SOC and Red Team (hosted): `scrollHeight === innerHeight`, dock/terminal/URL bar visible, strip not clickable with a modal open
1.2	OVF-02, OVF-03	ThreatWall BrandRow wraps or becomes icon buttons on small screens	S	No horizontal scroll at 320/375/430 px on `#/start` and `#/record`
1.3	OVF-04, OVF-05, OVF-06, OVF-12	DevSecOps: `wsbar` wraps or uses an overflow menu; segmented control wraps; chip wraps; editor scrolls horizontally (`overflow-x:auto`)	M	No horizontal page scroll at 320/375 px; every editor line reachable; Commit & Push reachable
1.4	OVF-07, OVF-10, OVF-09	Use `minmax(min(100%, Npx), 1fr)` for SOC, Red Team and site grids; `overflow-wrap:anywhere` on body copy; `min-width:0` on `.rdoc` text	S	No horizontal scroll at 320 px on `#/demo`, `/pilot.html`, `/sample-report.html`
1.5	OVF-08, OVF-11, OVF-13	Console top bar collapses below 600 px and panels default closed below 1100 px; AI Security unlock select wraps; xterm fit uses the host width	M	No horizontal scroll at 320/375 px; terminal lines fully visible at 375 px
1.6	OVF-15 to OVF-22	Browser-verify each risk, then fix the confirmed ones	M	Each item marked fixed or "not reproducible" with evidence
1.7	UX-18	Add "Best on a larger screen" notices where phone use is impractical (ThreatWall, DevSecOps, Red Team, SOC console); move Reset/Brief into a menu	S	Every action reachable at 375 px, or the notice explains the limitation

Phase 2: Core accessibility (about 6 days)

Order	IDs	Change	Effort	Acceptance criteria
2.1	A11Y-01	Editor and terminal: Esc then Tab (or Ctrl+M) leaves; indent only when no modifier key is held; document the key in visible help; enable xterm `screenReaderMode`	M	A keyboard user can Tab out of each editor and terminal; the help text names the key
2.2	A11Y-04	One shared accessible Dialog (role, `aria-modal`, initial focus, trap, Escape, restore focus, inert background) used by every app	M	Keyboard focus cannot leave an open dialog; Escape closes it; focus returns to the trigger
2.3	A11Y-02, A11Y-03	Red Team baseline: labels, buttons and links instead of clickable spans and rows, tab semantics, `aria-live` toasts, named icon buttons, `inert` closed drawers	L	axe has 0 critical/serious issues on all Red Team views; catalog → report can be completed by keyboard alone
2.4	A11Y-07, A11Y-14	Keyboard-operable map nodes (or an equivalent radio list); text alternatives for diagrams and the admin chart	M	The AI Security fix-layer step can be completed by keyboard; the accessibility tree lists diagram components

Order	IDs	Change	Effort	Acceptance criteria
2.5	A11Y-08, A11Y-10	Live regions for async states; per-route titles; focus to `h1` on navigation; an `h1` on every screen	M	Screen readers announce "Evaluating…" and then the score; every route has a unique title and an `h1`
2.6	A11Y-06, A11Y-09, A11Y-11 to A11Y-17	Border tokens ≥3:1, proper tab and radio patterns, underlined inline links, landmarks, reduced motion in the three labs, text instead of emoji meaning	M	axe AA clean across §10.1 pages in both themes

Phase 3: Workflow integrity (about 6 days)

Order	IDs	Change	Effort	Acceptance criteria
3.1	FUNC-04, FUNC-05	AI Security: "Finish anyway" after a failed re-test; persist the active attempt; hash routes per view; confirm on Exit	M	A failed re-test reaches Results marked failed; reload and Back resume the same phase
3.2	UX-09	AR: confirm Skip when text exists; save the draft per attempt; `beforeunload` guard	S	Typing then Skip asks first; reload restores the text and selection
3.3	FUNC-09	Namespace ThreatWall and DevSecOps progress by platform user (reuse AI Security's pattern); show an error rather than falling back to a shared key	M	User B never sees user A's progress on a shared browser
3.4	UX-07, FUNC-11, UX-16	InterviewPrep: opt-in or adjustable auto-submit with an announced countdown; handle all mic errors; non-dismissing error toasts; mic check before Start	M	Every recognition error shows a message and falls back to typing; the countdown is announced
3.5	UX-12, FUNC-07, FUNC-08, FUNC-06	Persist wizard step and report tab in the URL; poll for the PDF; retry on Profile; Turnstile failure message	M	Reload keeps step 2; the PDF button appears without a reload; with Turnstile blocked, each form shows an actionable message within 5 s
3.6	UX-10, UX-13, FUNC-12	Red Team: resume the active run; hide true severity; confirm report submission and show the score breakdown	M	Reload opens the workspace; severity hidden before submission; the report lists each penalty
3.7	UX-11, UX-14, FUNC-14 to FUNC-16, CONT-02	Clock copy or behaviour, mode lock, locked-URL side effect, disabled-after-complete, "Previous runs" list, human-readable error messages	M	Copy matches behaviour; no raw error codes visible

Phase 4: Design-system consolidation (about 5 days, can run in parallel with Phase 3)

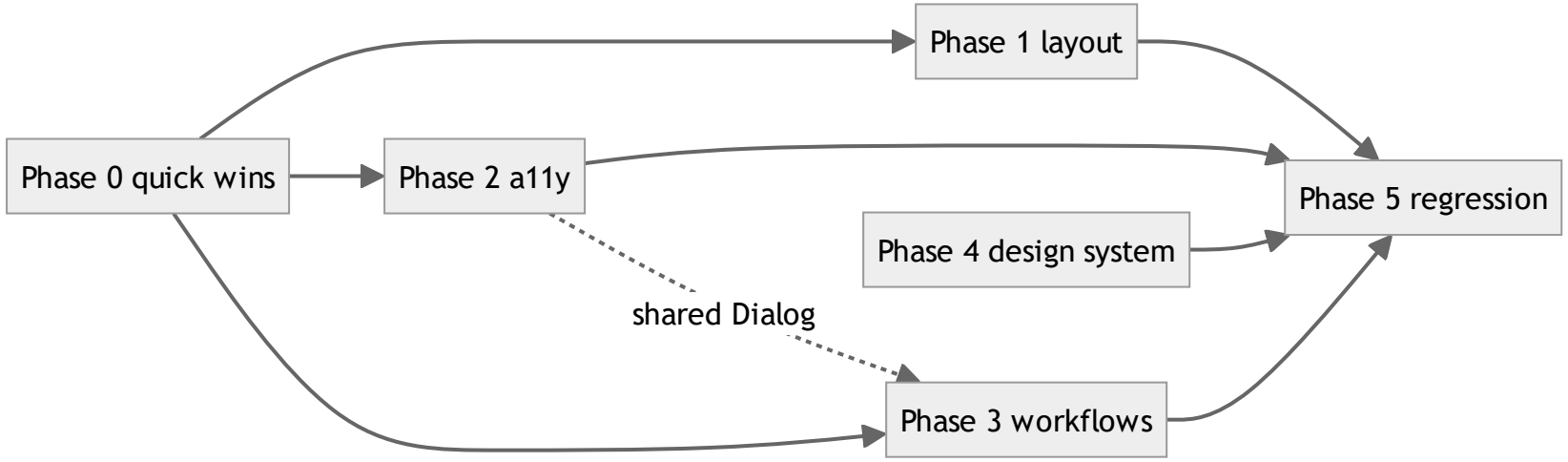
Order	IDs	Change	Effort	Acceptance criteria
4.1	VIS-03, VIS-04	Document the token set (`site.css` is the source of truth); align button height, container width, focus ring and body type across site, platform and apps	M	One documented button spec, used everywhere; one `--wrap` value

Order	IDs	Change	Effort	Acceptance criteria
4.2	VIS-05	Semantic colours: error, danger, low time and failed → `--bad`; product markers stay markers	S	No `--accent*` in error or danger rules; error colour ≥4.5:1 in both themes
4.3	VIS-07	Shared theme: the site writes and reads the `tr-theme` cookie; SOC, ThreatWall (training layer), DevSecOps and AI Security adopt `theme-init.js` + light tokens, **or** explicitly pin dark and say so; fix Red Team's dark-only colours	L	Toggling on threatready.io carries to every app; axe 0 contrast issues in light mode
4.4	VIS-06, VIS-08 to VIS-10, UX-08	Define the sample-report tokens; remove leftover palettes; re-sync Red Team `platform.css`; one brand per page	S–M	No undefined `var(--x)`; shared CSS files identical (checksum)

Phase 5: Regression automation (about 3 days)

Implement §12.3 and add it to the existing `e2e` package. Acceptance: the suite runs in CI or locally in under 10 minutes and fails on any regression of the items fixed above.

Dependencies



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15. Open questions and product decisions

1. **Trial scope (FUNC-03).** Should a trial session be able to use any product, or is the trial only a guided look around?
2. **Supported devices.** Are phones a supported target for the dense labs (TR-CLOUD-001, Red Team, SOC console, DevSecOps, ThreatWall), or should they show a "use a larger screen" notice? This decides whether OVF fixes there aim for full use or graceful degradation.
3. **Light theme.** Should every lab follow the platform theme, or may some (the AWS-style console, the ThreatWall appliance) stay pinned on purpose? If pinned, the training layers should still match the platform.
4. **Lab Simulations status.** Is it "in development" (site) or "Live" (onboard, README)?
5. **SOC stage labels (UX-20).** Is Closure done by Tier 1 on purpose?
6. **Inspect costs time (UX-11).** Is it intended that viewing evidence moves the intruder clock?
7. **Red Team severity (UX-13) and AI Security finish rules (FUNC-04).** Pedagogical choices to confirm with the content owners.
8. **Legal content (CONT-01).** Registered address and grievance officer details are needed from the business.
9. **Tab-switch guard (CONT-08).** Should it be advisory only, or recorded on the server?

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16. Appendix: evidence index and how to reproduce

16.1 Screenshots (`docs/audit-evidence/`)

File	Shows	Findings
`labs-console-1440x900.png`	TR-CLOUD-001 briefing; lab-nav strip undimmed above the modal; duplicate brand	OVF-01, UX-08
`labs-console-375x812.png`, `labs-console-320x568.png`	Console top bar crowding / overflow on phones	OVF-08
`labs-track-blue-1440x900.png`	"No labs in this track yet"	UX-01
`tw-home-375x812.png`, `tw-record-375x812.png`	ThreatWall header overflow	OVF-02, OVF-03
`flow-threatwall-lab-375x812.png`	ThreatWall lab on a phone (truncated URL bar, tooltip overlap, task panel below)	OVF-14
`dso-welcome-320x568.png`, `dso-dashboard-375x812.png`, `flow-devsecops-lab0-375x812.png`	DevSecOps overflow and editor clipping	OVF-04, OVF-05, OVF-06, OVF-12
`soc-demo-320x568.png`	SOC demo intro overflow	OVF-07
`flow-soc-case-after-4s-375x812.png`, `flow-soc-case-after-4s-1366x768.png`	SOC console on phone and laptop	UX-03, UX-20
`flow-aisec-mission-control-375x812.png`	Unlock-mode select overflow	OVF-11, UX-15
`flow-redteam-briefing-modal-1366x768.png`	Red Team briefing modal	A11Y-02
`flow-redteam-tab-terminal-375x812.png`	Terminal clipped on a phone	OVF-13
`light-redteam-proxy.png`	Red Team light theme; tools pushed below the fold	UX-02, VIS-07
`site-sample-report-375x812.png`, `site-pilot-320x568.png`	Marketing overflow	OVF-10, OVF-09

16.2 Key measurements

- **TR-CLOUD-001 with the strip, 1366×768:**
  - document 1,528 px
  - `.app` 40→1,528
  - `.main` 108→1,318 (`overflow-y:auto`)

    - `.side` 108→1,528 (`overflow-y:visible`)
    - `.dock` 1,318→1,528
- **TR-CLOUD-001 with the strip, 768×1024:** document 1,703 px.
- **Strip injected, 1366×768:**
    - ThreatWall lab: 768 → 808 px
    - DevSecOps workspace: 768 → 808 px
    - AI Security workspace: 924 → 964 px
- **DevSecOps editor at 375 px:** `scrollWidth` 476 / `clientWidth` 354, `overflow-x:hidden`, no label. After Tab, `activeElement` is still `TEXTAREA`.
- **Red Team at 375 px:**
    - drawers at x −322…0 and 375…698, `visibility:visible`, not inert, with 3 and 2 focusables
    - Reset and Brief `offsetParent === null`
    - xterm screen 384 px vs host 341 px
    - after Tab, focus is still in `xterm-helper-textarea`
- **AI Security:** the reload during a scenario returns to "Mission Control — Welcome back".
- **Unknown routes:** SOC `#/level/does-not-exist` → `h1` "SOC simulation"; DevSecOps `#/lab/nope` → `h1` "Welcome to the ThreatReady DevSecOps Lab".

16.3 Environment and reproduction

- **Host:** Windows 11, Node 26.10, Google Chrome (stable). Docker was not available, so PostgreSQL-backed apps were blocked.
- **Servers** (each from its own folder; nothing in the repository was changed):
    - static site: `python -m http.server 8090 --directory Landing_Page-main`
    - ThreatWall: `npx vite --port 5181`
    - DevSecOps: `npx vite --port 5183`
    - AI Security: `npx vite --port 5182`
    - SOC: `npm run start:dev` + `npx vite --config client/vite.config.js`
    - Red Team: `npx tsx server/index.ts` + `npx vite`
    - Labs: `npx tsx server/index.ts` with `IDENTITY_URL` pointed at a local stub of `/api/internal/identity/session` that returns one test user, plus `npx vite`
- **Tooling:** `playwright-core` 1.53.2 and `axe-core` 4.10.3, run from a temporary folder outside the repository.
    - Sweep: the 8 viewports in §9, `networkidle` + settle time, a horizontal-overflow offender scan, axe at 1440×900 (and 375×812 on entry pages) with `bypassCSP` so axe can inject on the marketing site's strict CSP.
- **Lab data written during testing** stayed in git-ignored locations (`soc/data/`, `red_team_lab/web/data/`) or in the temporary folder.
- **Source reviews:** five parallel read-only reviews (onboard + site, InterviewPrep, Labs + AR, SOC + Red Team, ThreatWall + DevSecOps + AI Security). Every SC/SI finding above cites `path:line` from those reviews or from my own checks.

17. Fix status (23 Oct 2026)

The fixes are in the working tree on branch `ui-ux-audit-fixes`. Nothing is committed or pushed.

17.1 Summary

Result	Count	IDs
Fixed	86	everything not listed below
Partly fixed	6	FUNC-07, OVF-21, A11Y-04 (Labs console), A11Y-08 (by design), VIS-09, CONT-04
Left open on purpose	4	CONT-01 (needs business details), UX-20 (product question), VIS-04 (design decision), CONS-02 (expected browser log)

Verified after the fixes:

Check	Before	After
Horizontal overflow: marketing site (17 pages × 8 viewports)	3 failures	**0**
Horizontal overflow: runnable apps standalone (21 pages/states × 8 viewports)	16 failures (+1 in the AI Security walkthrough)	**0**
Horizontal overflow: the five hosted labs under `/labs/<slug>/` with the real strip (×8 viewports)	not measurable	**0**
All runnable pages together (43 pages/states × 8 viewports, final run)	19 failures	**344 of 344 pass**
Lab consoles taller than the viewport under the strip (TR-CLOUD-001, SOC, DevSecOps, ThreatWall at 1366×768)	TR-CLOUD-001 1,528 px; ThreatWall/DevSecOps 808 px	**768 px, exactly the viewport**
axe issues on the 43 runnable pages (WCAG 2.1 AA + best practice)	11 rule types, 241 colour-contrast nodes	**0 critical, serious or minor**. 3 moderate "region" notes remain (skip links outside landmarks). The Red Team workspace is clean on all 5 tool tabs, desktop and phone, dark and light
Page errors / console errors	React duplicate-key warning (ThreatWall)	**none**, apart from the expected 401 from Red Team's signed-out check
Typecheck and build, all 10 apps	—	**pass**
Unit tests	—	**pass**, except 2 that also fail on the original code (below)

Tests that also fail on the original code, so not caused by these changes:

- `InterviewPrep/apps/web/test/avatar-asset.test.ts` needs the licensed `interviewer.glb`, which is not in the repository.
- `soc/tests/variants.test.mjs` ("a perfect run typed with the renamed values…") fails the same way on commit `93cc4bf`, checked in a clean worktree.

Still not verified in a browser: onboard, InterviewPrep and Attack Reasoning (they need PostgreSQL and LLM services). Their fixes are checked by typecheck, unit tests and build only.

17.2 Status per finding

Key:

- **RC** = re-checked in a browser after the fix.
- **SC** = fixed in source and confirmed by typecheck, tests and build, but not exercised in a browser.

ID	Status	Verified	What changed
FUNC-01	Fixed	SC + link scan	CTAs go to `onboard.threatready.io/` and `/?mode=signup`; `404.html` loop removed; the site checker fails on any `threatready.io/app` link
FUNC-02	Fixed	**RC**	SOC stream and downloads use the app base; the hosted stream connects at `/labs/soc/api/cases/:id/stream` (200)
FUNC-03	Fixed (copy)	SC	Guest-session copy in onboard and the site FAQ; access rules unchanged
FUNC-04	Fixed	SC	AI Security: "Finish anyway & see results" after a failed re-test; the attempt is recorded as not passed
FUNC-05	Fixed	**RC**	AI Security attempt saved per user; hash routes; exit confirmation. A reload resumes the same scenario
FUNC-06	Fixed	SC	Turnstile shows a "Verifying…" state, plus an error and retry after 5 s
FUNC-07	Partly fixed	SC	The report polls for 2 minutes and says "Preparing PDF…". The API cannot tell "pending" from "disabled"
FUNC-08	Fixed	SC	Retry on the Profile overview and on the Home "Continue" strip
FUNC-09	Fixed	**RC**	ThreatWall and DevSecOps progress is keyed per user (`…:u:<id>`) and old shared data is migrated once. If the platform API doesn't answer within 4 s, those two fall back to the old shared key. AI Security is stricter: it switches to session-only storage and never uses a shared key
FUNC-10	Fixed	SC	The confirm field no longer shows the email as a placeholder; server errors appear at form level
FUNC-11	Fixed	SC	Every mic error shows a message and switches to typing; optional mic test
FUNC-12	Fixed	SC	Red Team report submission asks for confirmation and shows a score breakdown

ID	Status	Verified	What changed
FUNC-13	Fixed	SC	AR `/scenarios`: dead button replaced by a "Planned" tag; stale copy reworded
FUNC-14	Fixed	SC	A locked DevSecOps lab shows only the locked message and saves no state
FUNC-15	Fixed	SC	Completed TR-CLOUD-001 runs are read-only, with a banner
FUNC-16	Fixed (as a modal)	SC	"Previous runs…" lists archived runs with their read-only report
FUNC-17	Fixed	**RC**	Red Team hides Sign out when hosted, as SOC does
OVF-01	Fixed	**RC**	New lab-strip height contract (`--tr-labnav-h`); consoles fit the viewport; modals cover the strip
OVF-02, OVF-03	Fixed	**RC**	ThreatWall header wraps
OVF-04, OVF-05, OVF-06	Fixed	**RC**	DevSecOps bar, mode switch and chip wrap; locked cards and their buttons wrap
OVF-07	Fixed	**RC**	`minmax(min(100%,N),1fr)` grids in SOC and Red Team
OVF-08	Fixed	**RC**	Console top bar collapses below 600 px; panels closed by default below 1100 px
OVF-09, OVF-10	Fixed	**RC**	Pilot list and sample-report cards wrap
OVF-11	Fixed	**RC**	AI Security unlock mode moved into Settings, at full width
OVF-12	Fixed	**RC**	DevSecOps editor scrolls sideways (scroll width 556 px in a 375 px view); no clipped code
OVF-13	Fixed	**RC**	xterm fits its host (336 px in 341 px)
OVF-14	Fixed	**RC** (layout)	ThreatWall URL bar and Lab Tools on phones
OVF-15, OVF-16, OVF-17	Fixed	SC	InterviewPrep sticky offsets and word wrapping; onboard range picker scrolls
OVF-18, OVF-19, OVF-20, OVF-22	Fixed	RC (18), SC	Red Team browser bar wraps; AI Security results grid; ThreatWall toasts; AR level selector
OVF-21	Partly fixed	SC	Edge labels readable on hover; a text list of the diagram is added. The diagram itself stays small on phones
A11Y-01	Fixed	**RC**	Esc then Tab leaves the DevSecOps editor and terminal and the Red Team terminal; visible hint
A11Y-02	Fixed	**RC**	Red Team: labels, buttons, tab pattern, dialogs, live toasts, landmarks. axe clean on every tool tab
A11Y-03	Fixed	**RC**	Closed drawers are `inert` and `aria-hidden`
A11Y-04	Fixed; partly in Labs	SC	Focus trap, Escape and focus restore in every app's dialogs. Labs console dialogs don't make the page behind inert
A11Y-05	Fixed	**RC** (axe)	Violet text uses `#A59BFF`; darker primary hovers

ID	Status	Verified	What changed
A11Y-06	Fixed	SC	Input borders ≥3:1 in onboard, InterviewPrep and the site
A11Y-07	Fixed	SC	Keyboard-operable AI Security map and Labs architecture nodes
A11Y-08	Fixed; one exception by design	SC	Live regions for async states. InterviewPrep's live dictation transcript is deliberately not read aloud, so it doesn't talk over the speaker
A11Y-09 to A11Y-17	Fixed	RC (11, 12 via axe) / SC	Tab and radio patterns, titles and `h1`s, underlined links, landmarks, skip link, diagram and chart alternatives, non-colour status, reduced motion, emoji
UX-01	Fixed	**RC**	The Blue Team track lists SOC Simulation (`docs/audit-evidence/after-labs-track-blue-1440x900.png`)
UX-02	Fixed	SC	Red Team beginner guidance can be collapsed, and the choice is remembered
UX-03	Fixed	RC (layout)	SOC narrow layout: one page scroll and jump links
UX-04	Fixed	SC	AR diagram and question side by side at ≥1200 px
UX-05, UX-06, UX-07, UX-12, UX-16	Fixed	SC	InterviewPrep: options behind a toggle, smaller hero heading, auto-submit opt-in and announced, state survives reload, persistent error toasts
UX-08	Fixed	SC	No duplicate brand under the strip (console, ThreatWall)
UX-09	Fixed	SC	AR Skip asks for confirmation; drafts saved per attempt
UX-10	Fixed	**RC**	Red Team reload resumes the active run
UX-11	Fixed (copy)	SC	Briefing, help and tooltip say that inspecting also advances the clock
UX-13	Fixed	SC	Red Team severity is withheld (server side) until the learner reports it
UX-14	Fixed	SC	DevSecOps mode is locked per attempt; the timer counts active time only
UX-15	Fixed	SC	Unlock mode is in Settings with an explanation
UX-17	Fixed	**RC**	Not-found pages in SOC, DevSecOps and Labs
UX-18	Fixed	**RC**	Red Team "More" menu (Brief, Reset); DevSecOps file picker and keyboard/touch splitter
UX-19	Fixed	SC	ThreatWall idle timeout resets on activity and warns a minute before logout
UX-20	**Open**	—	Product question (SOC "Tier 1 · Closure")
VIS-01, VIS-02	Fixed	SC	AR score bars render; class collision removed
VIS-03	Fixed	SC	App buttons are 42 px (onboard, InterviewPrep); the marketing site keeps 44 px

ID	Status	Verified	What changed
VIS-04	**Open (decision)**	—	The marketing site keeps its wider container and larger body type; the app shell keeps 1280 px / 16 px
VIS-05	Fixed	SC	InterviewPrep errors, low time and failing scores use the red family, with icons
VIS-06	Fixed	**RC** (axe)	Sample-report colours tokenised; contrast passes
VIS-07	Fixed	RC (Red Team light, axe clean) / SC	Site shares the `tr-theme` cookie; Red Team readable in light theme (terminal stays dark by design); SOC, ThreatWall, DevSecOps and AI Security declared dark-only
VIS-08	Fixed	SC	SOC teal and pink leftovers removed; ThreatWall training teal is now `#3fb6a8`. ThreatWall keeps IBM Plex for the appliance look
VIS-09	Partly fixed	—	Hardcoded colours replaced in every file touched; there was no repo-wide token sweep
VIS-10	Fixed	SC	Labs marker on the track page
CONT-01	**Open**	—	Registered address and grievance officer must come from the business. `terms.html:171-172` still describes the old free trial, for legal to decide
CONT-02	Fixed	SC	Plain-language error titles in the console
CONT-03	Fixed	SC	Lab Simulations shown as live across the site
CONT-04	Partly fixed	SC	AR flags and InterviewPrep levels are explained. No glossary was added for SOC, AI Security or ThreatWall jargon
CONT-05 to CONT-08	Fixed	SC	SOC reset note follows the setting; clearer labels; SOC page title; neutral tab-guard copy
CONS-01	Fixed	**RC**	ThreatWall duplicate step id renamed (no console warning); a test now enforces unique ids
CONS-02	**Open (expected)**	—	The browser logs the expected 401 from the signed-out session check

17.3 Shared changes and new pieces

- `platform.css` is identical again in all five copies: onboard, InterviewPrep, Labs, Attack Reasoning and Red Team (Red Team had drifted).
  - Dark `--accent-hover` is `#5b4bf0`: white text 5.7:1, previously 3.76:1.
  - `.tr-loading` respects the strip height.
  - `.tr-sr` is pinned top-left, so hidden text can't widen the page.
- `lab-nav.js` is a fixed 40 px strip that publishes `--tr-labnav-h`, pads `body`, sits below modals and hides in print.
- `browser-notice.js` (4 copies): Firefox is no longer recommended for voice interviews.
- `e2e/ui-audit.mjs` (`npm run ui-audit` in `e2e/`) is the regression sweep from §12.3: 8 viewports, no sideways scroll, consoles fit under the strip, axe WCAG 2.1 AA, and no `threatready.io/app` links.

- **Other bugs fixed along the way:**
  - AR "Next question" carried the previous attempt's answer (missing React key).
  - AI Security guardrail switches could not be clicked.
  - The TR-CLOUD-001 Help dialog closed on any click, and the "/" search shortcut didn't work.
  - Hidden live-region text added extra page height or width in DevSecOps and SOC.

17.4 After-fix evidence (`docs/audit-evidence/after-*.png`)

File	Shows
`after-labs-console-1440x900.png`	TR-CLOUD-001 under the strip: no duplicate brand; the modal backdrop covers the strip
`after-labs-console-375x812.png`	Console top bar collapsed on a phone
`after-labs-track-blue-1440x900.png`	The Blue Team track lists SOC Simulation
`after-tw-home-375x812.png` , `after-tw-record-375x812.png`	ThreatWall header wraps; small-screen notice
`after-dso-welcome-320x568.png` , `after-dso-dashboard-375x812.png` , `after-hosted-devsecops-375x812.png`	DevSecOps without sideways scrolling, standalone and hosted
`after-soc-demo-320x568.png`	SOC demo intro fits at 320 px
`after-hosted-redteam-375x812.png`	Red Team hosted on a phone
`after-site-sample-report-375x812.png` , `after-site-pilot-320x568.png`	Marketing pages fit

17.5 Still to do

1. Business and legal input: CONT-01 and the old trial clause in `terms.html` .
2. Product decisions: UX-20; VIS-04; whether the Red Team terminal should follow light theme; the FUNC-04 rule that a failed re-test never counts as a pass.
3. Browser checks for onboard, InterviewPrep and Attack Reasoning once PostgreSQL is available. Run `e2e/ui-audit.mjs` and `e2e/cross-browser.mjs` against the full local stack.
4. Optional follow-ups:
   - An API `pdf.status` value (pending / ready / disabled) to finish FUNC-07.
   - `inert` on the page behind Labs console dialogs.
   - A Web Worker with a timeout for DevSecOps learner code (an infinite loop can still freeze the tab).
   - A glossary for SOC, AI Security and ThreatWall jargon.

- Removing `body{overflow-x:hidden}` from `site.css` after a browser pass.

Evidence gallery

ThreatReady | DevSecOps Lab · AeroShop simulation

Mode Guided Standard Expert

My DevSecOps labs

Each lab is a working repository with its own terminal, pipeline and staging environment. Your work is saved in this browser.

0/8 labs completed

0 points earned

Start: Fix a red build →

LEVEL 0 **Foundations** — Git, CI pipelines and tests as gates

Foundations · CI/CD ~15 min

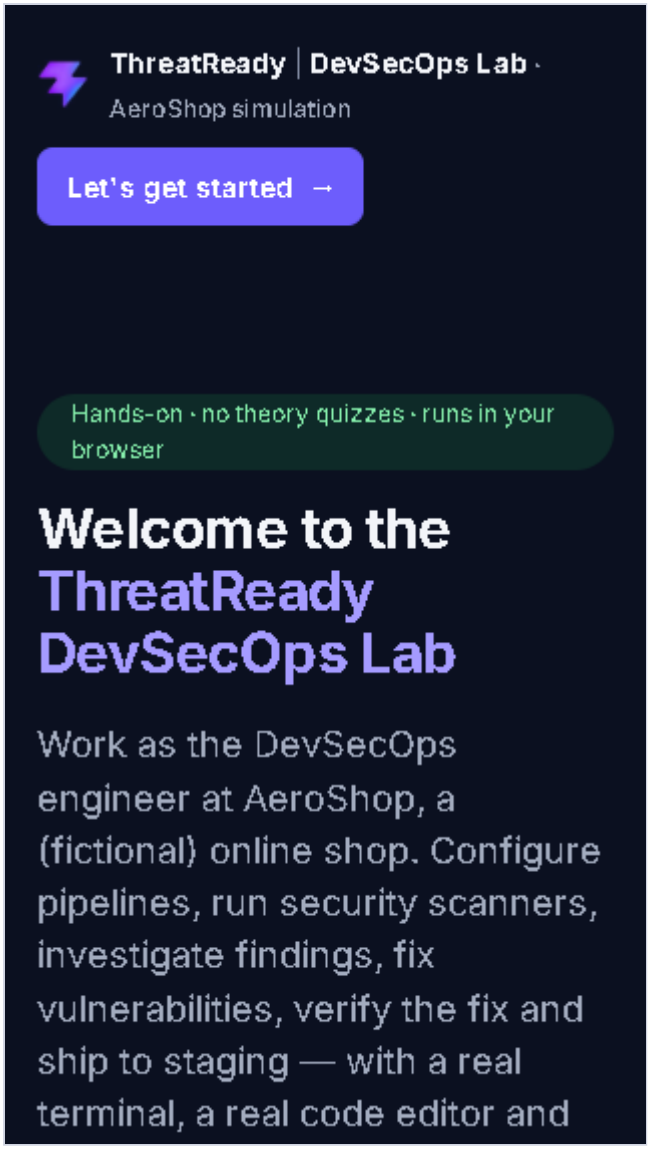
Fix a red build

Read a failing pipeline, reproduce it locally, fix the code and ship a green build through Git.

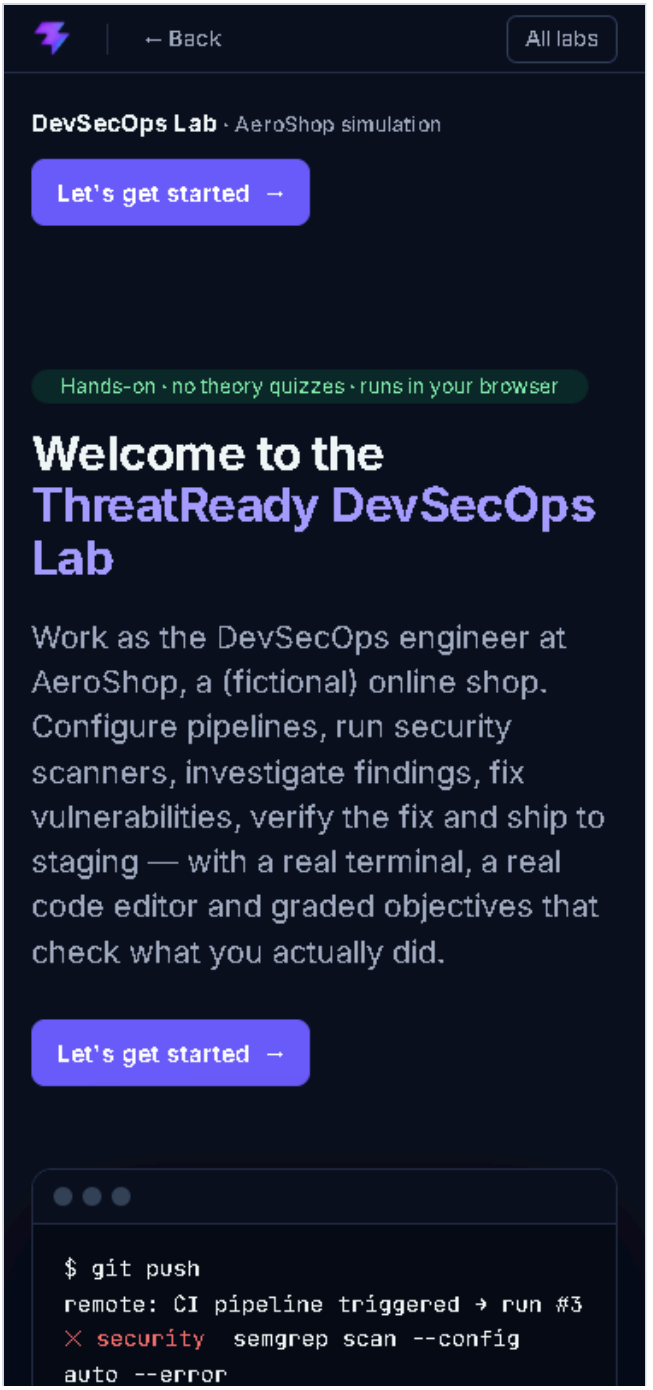
GitHub Actions git npm Jest

5 objectives + bonus Start lab

LEVEL 1 **Shift-left basics** — Secrets and static analysis

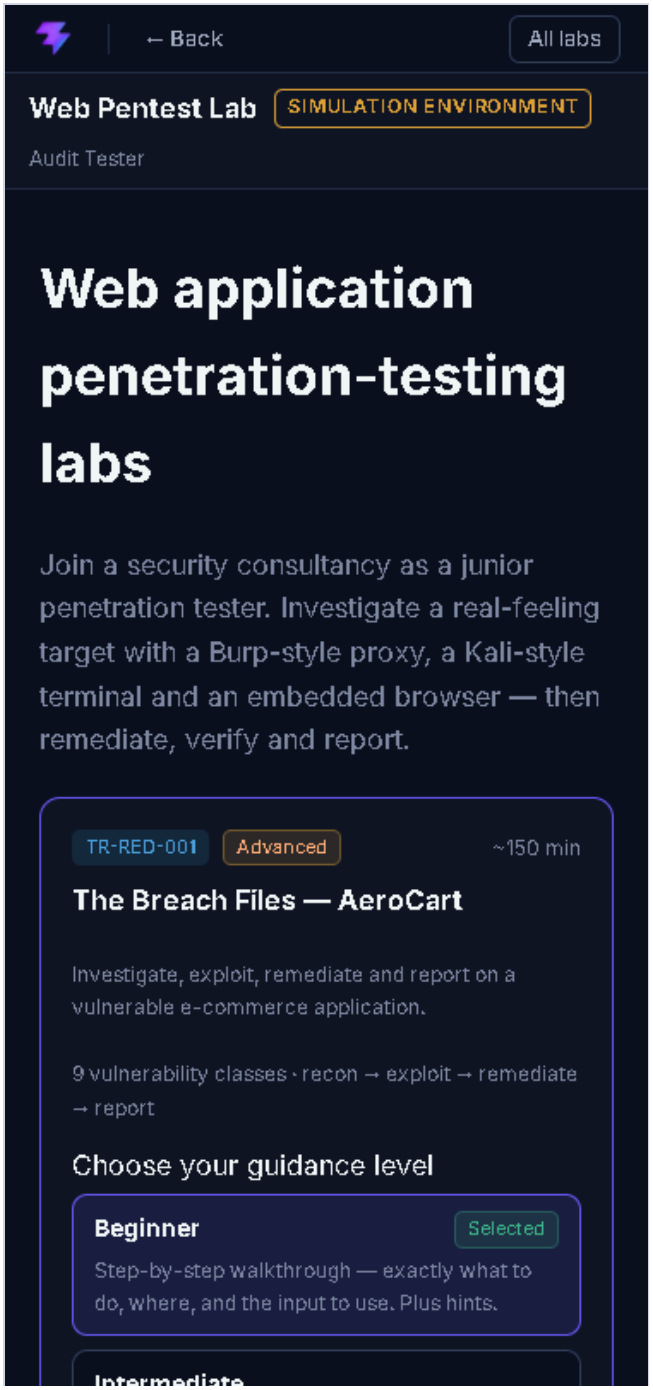


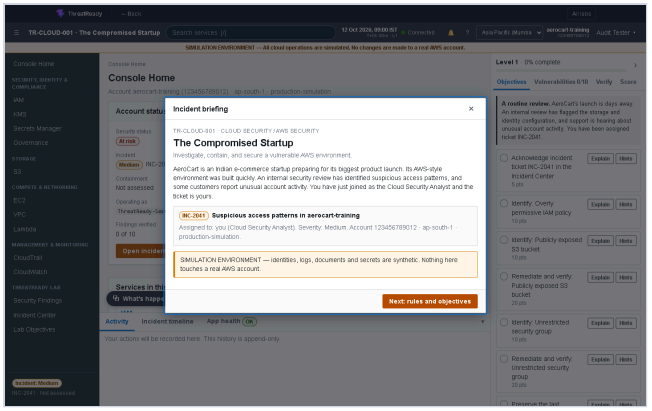
after-dso-welcome-320x568.png



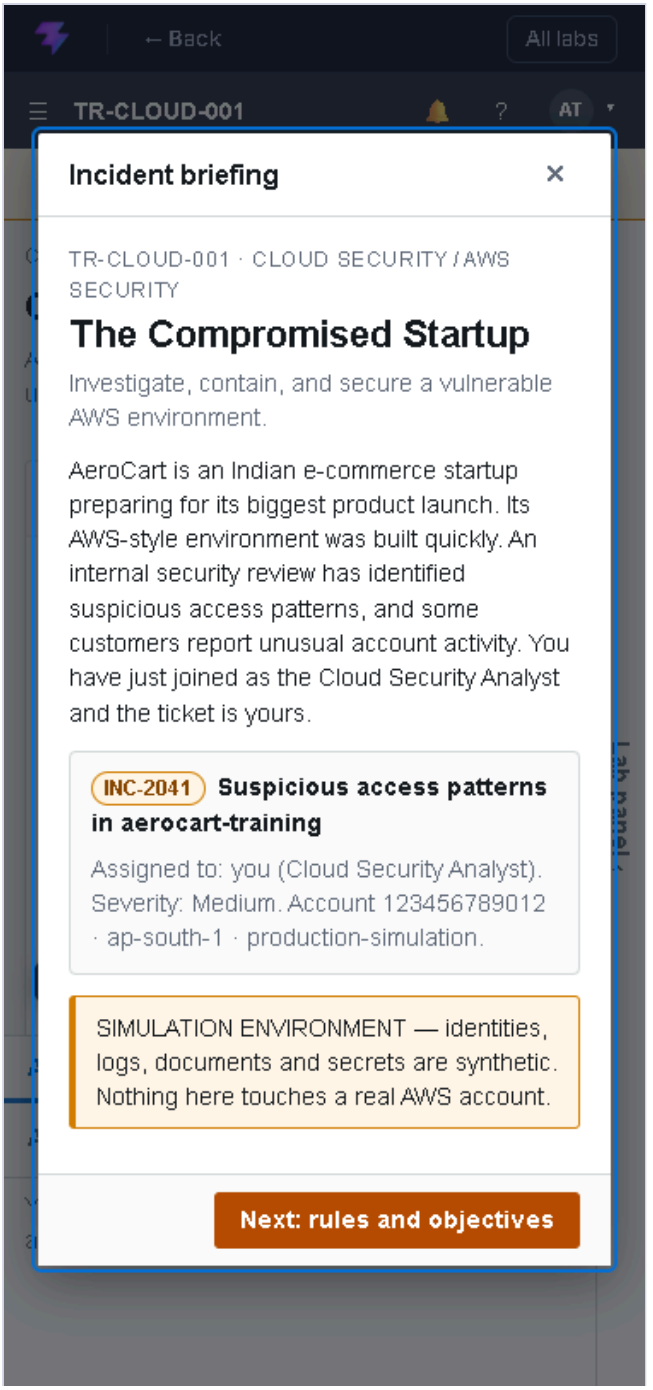
after-dso-dashboard-375x812.png

after-hosted-devsecops-375x812.png



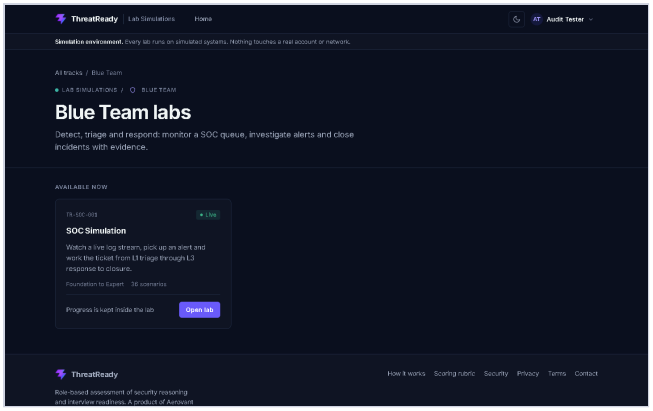


after-labs-console-1440x900.png

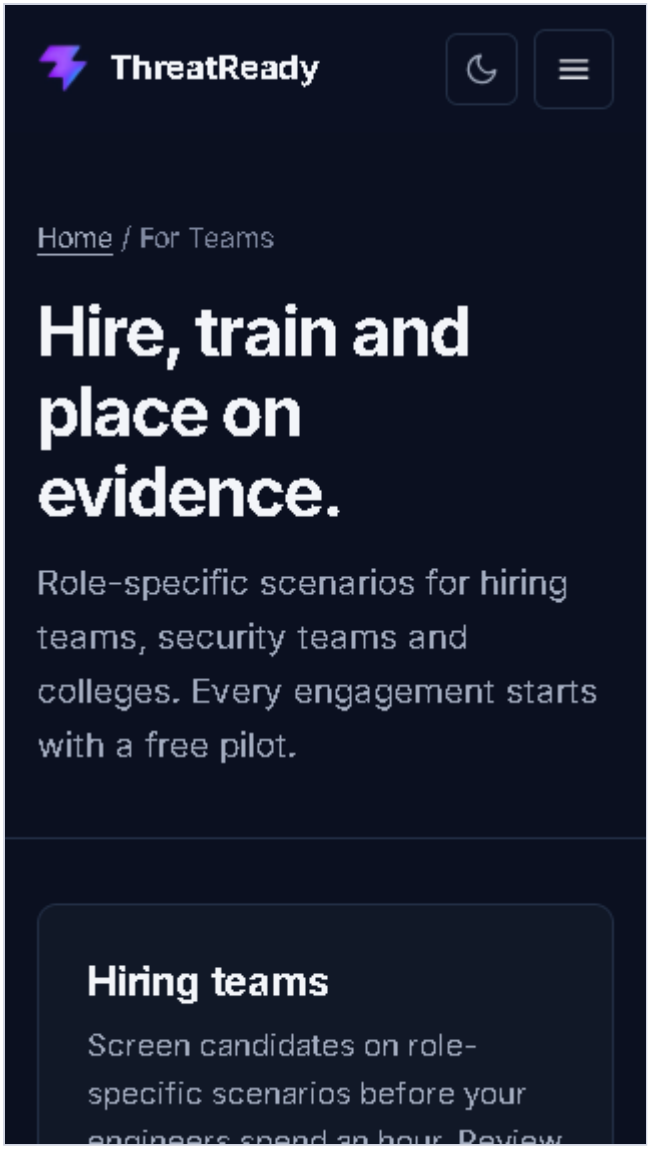


after-hosted-redteam-375x812.png

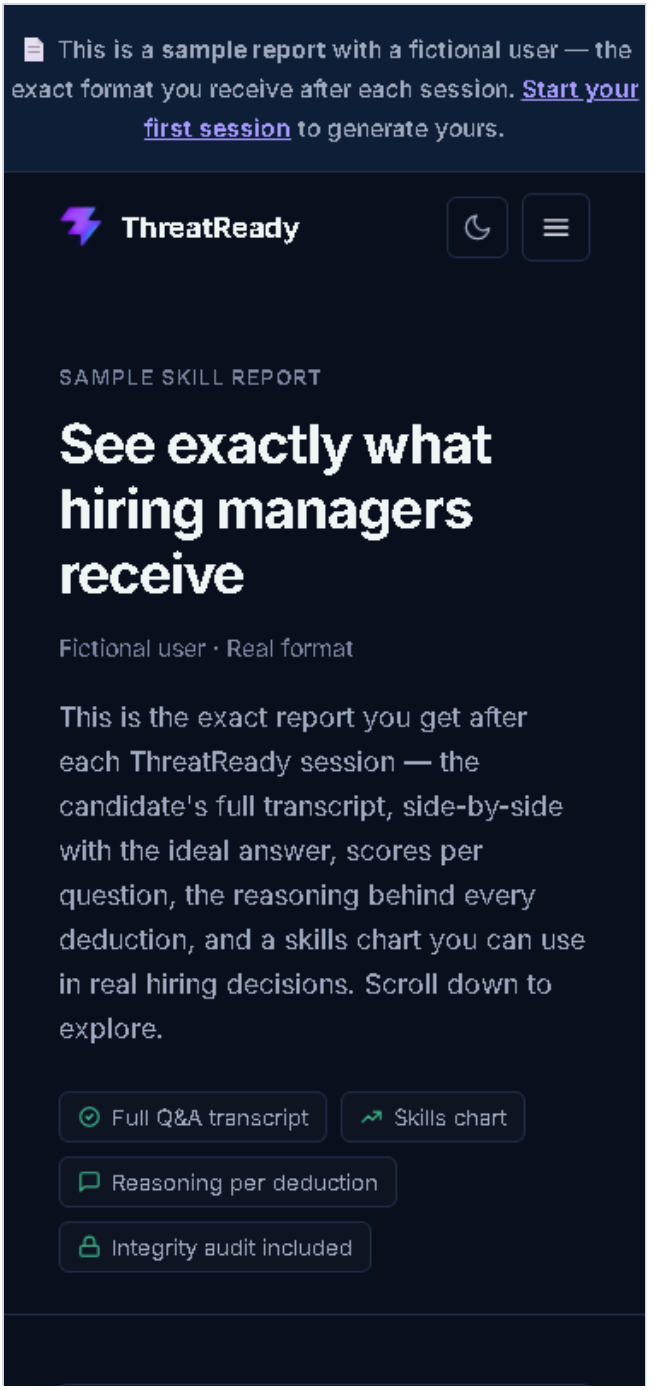
after-labs-console-375x812.png



after-labs-track-blue-1440x900.png



after-site-pilot-320x568.png



after-site-sample-report-375x812.png

Home

Guided practice, sample scenario, simulated data

Newbie Demo Mode

Learn how a SOC analyst investigates security events, reads logs, collects evidence, validates findings, and completes an incident through a guided, step-by-step demonstration.

Demo ticket analyst **dev.analyst**

Before you start

- A Security Operations Centre (SOC) watches an organisation's logs and reacts when something looks wrong.
- The work is split into three tiers. Tier 1 (L1) looks at every alert and decides whether it is real. Tier 2 (L2) works out exactly what happened. Tier 3 (L3) stops the attack and cleans up. In this demo you play all three, one after the other, on the same ticket.

after-soc-demo-320x568.png

AEROVANT TECHNOLOGIES | THREATREADY

Learning notebook | Training record

ThreatReady Firewall works best on a larger screen (a tablet in landscape, a laptop or a desktop). On a phone the task panel sits below the lab and wide tables scroll sideways.

ThreatReady Firewall

ThreatReady Firewall by Aerovant Technologies: hands-on firewall administration training from first cable to incident containment. You configure a fictional firewall, ThreatWall, in a lab that runs entirely in your browser: cabling, interfaces, rules, VPNs, high availability and a final incident.

Learner name — Shown on your training record

CONTINUE WHERE YOU LEFT OFF

Task 0.1 — Cable the PC and get an address

Level 0 — Access · ~5 min · step 1 of 4

Continue →

LEARNING NOTEBOOK

Every task explained, with flow diagrams

AEROVANT TECHNOLOGIES | THREATREADY

Home | Learning notebook

Back to lab

Training record

Learner name not set · ThreatReady Firewall · generated 10/23/2026, 2:28:23 PM · every check re-run against the current lab configuration

In progress: 0 of 30 tasks verified

Print or save as PDF

Download my record (JSON)

SUMMARY

0/30
tasks verified now

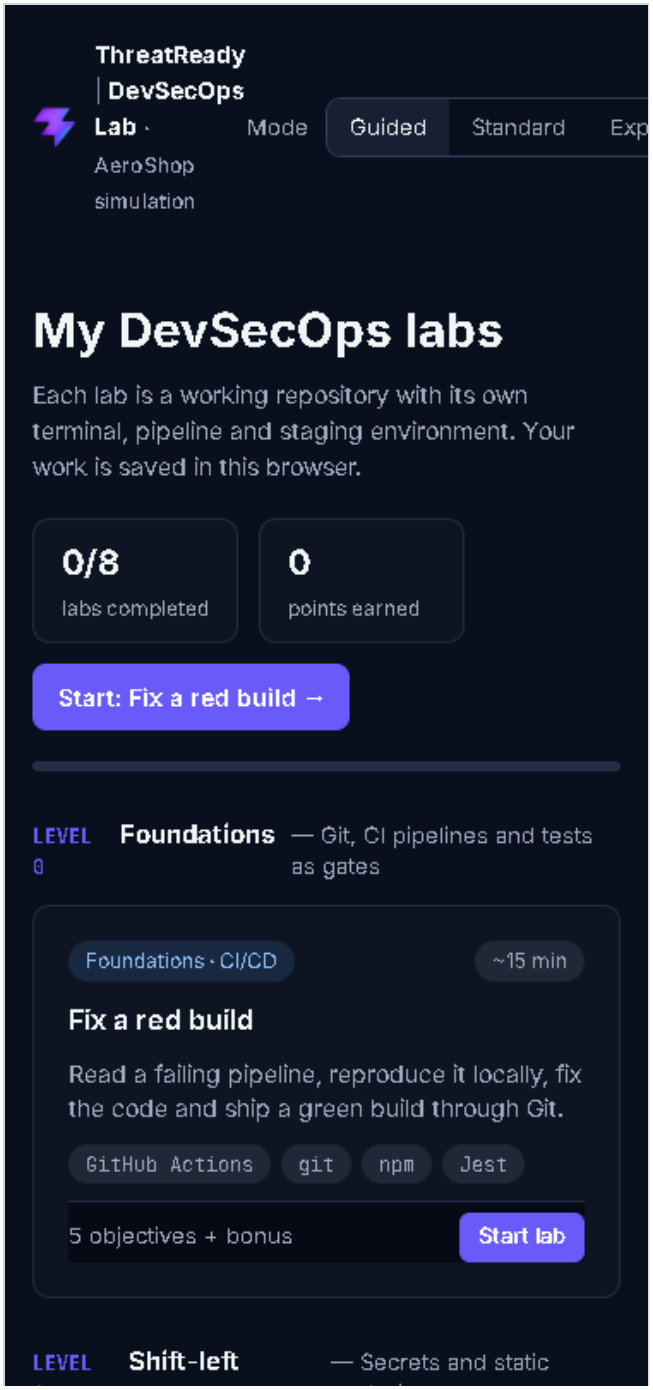
0
failed attempts recorded

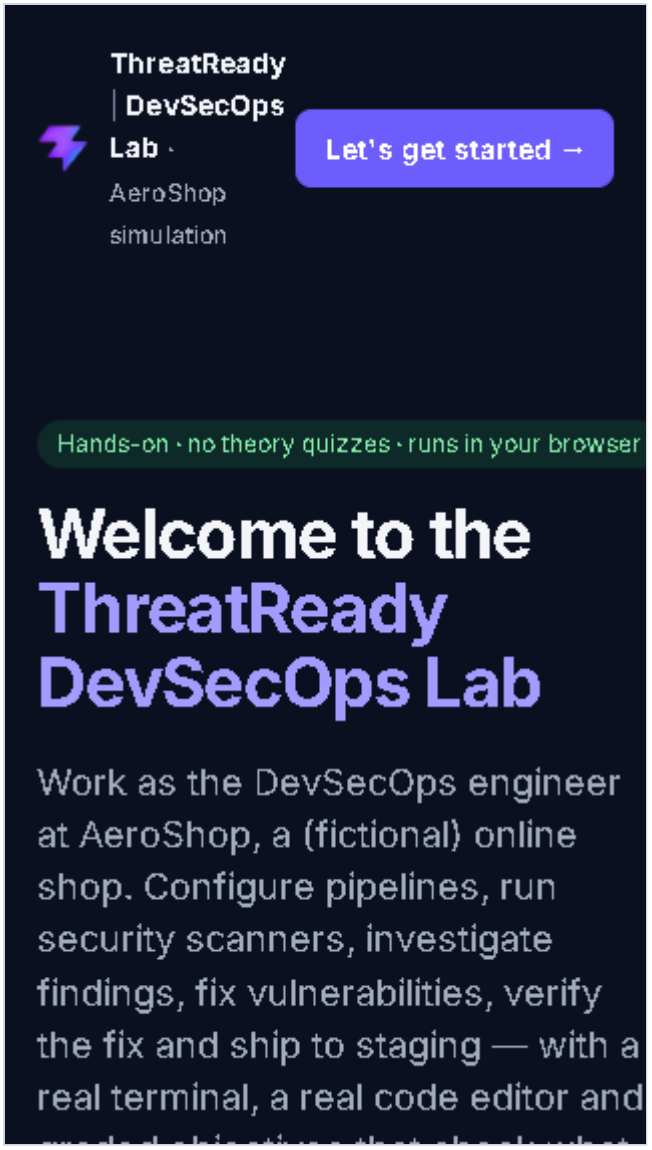
0
need re-verify

—
first to last completed task

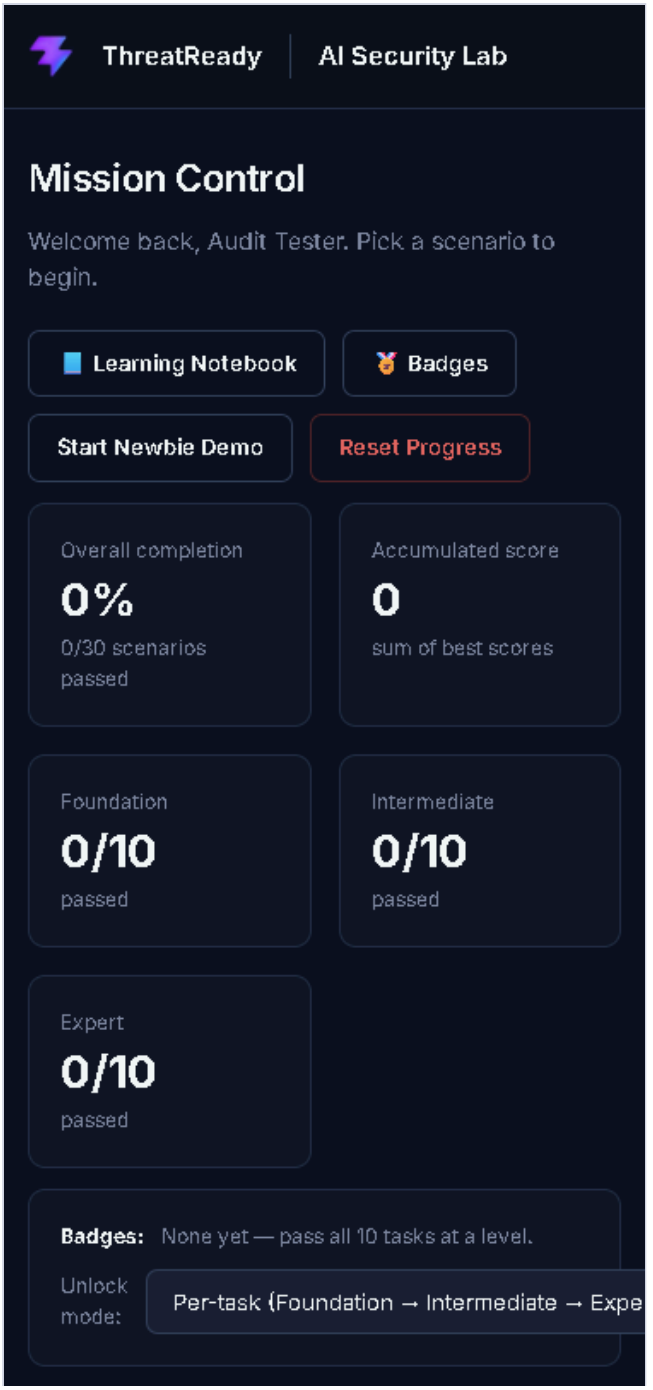
after-tw-home-375x812.png

after-tw-record-375x812.png



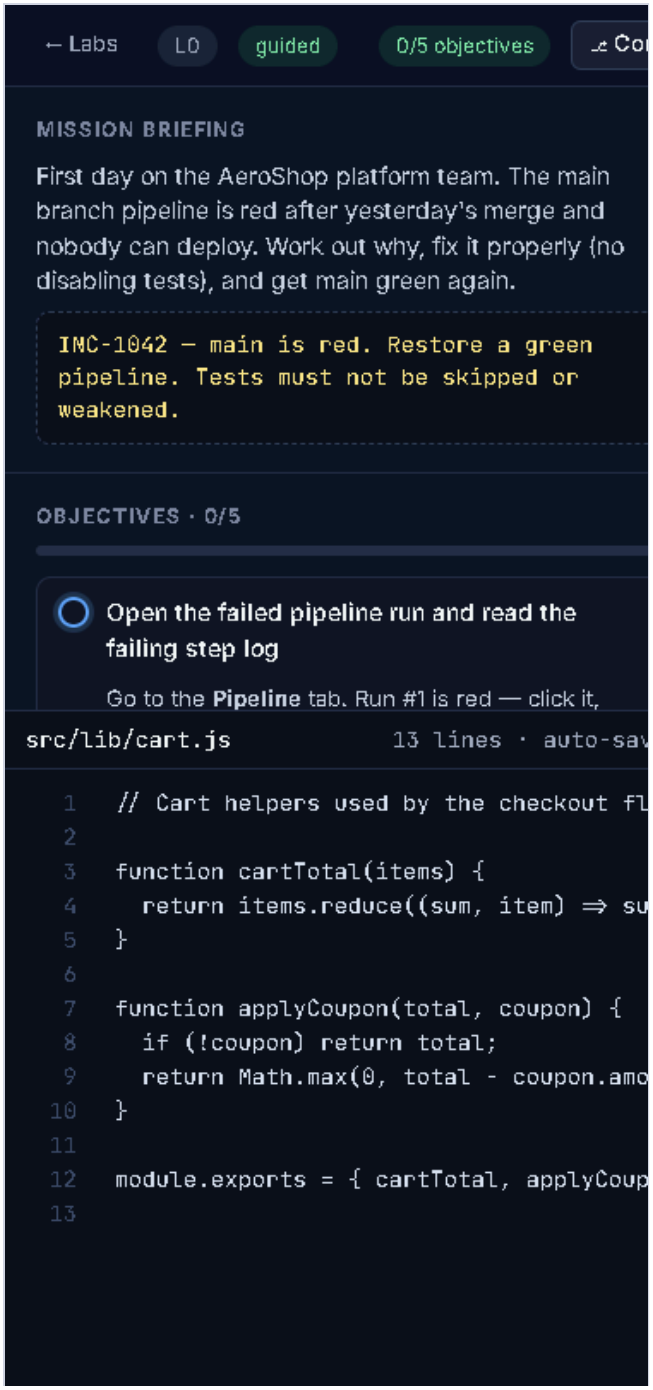


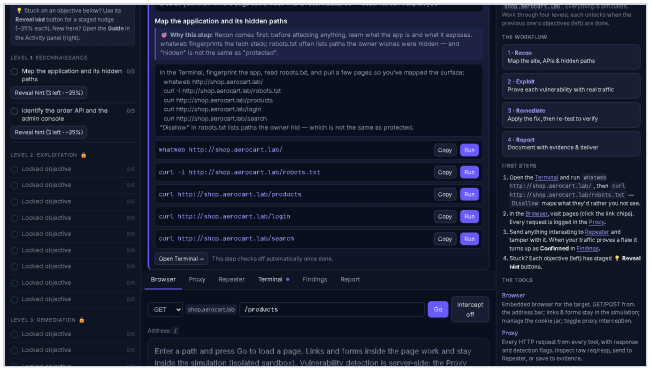
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dso-dashboard-375x812.png

flow-aisec-mission-control-375x812.png



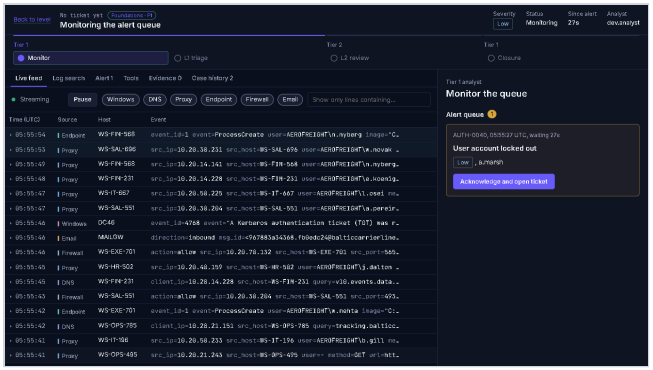


flow-redteam-briefing-modal-1366x768.png

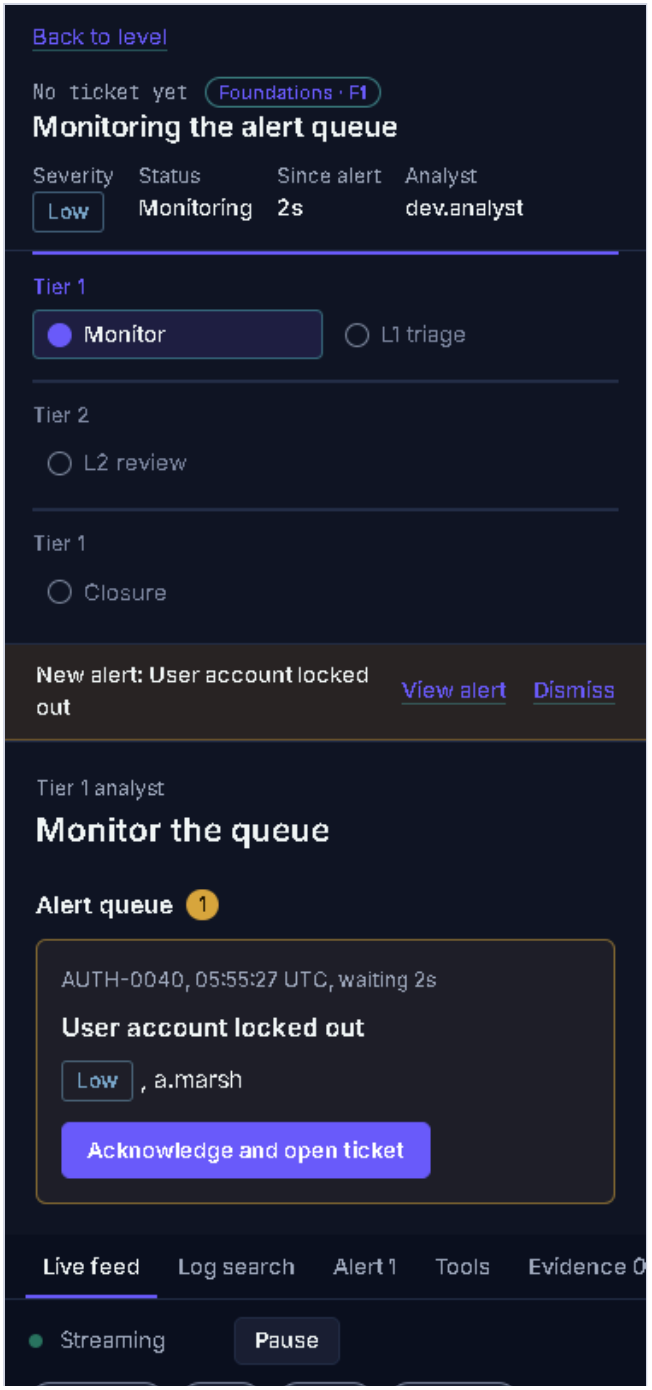


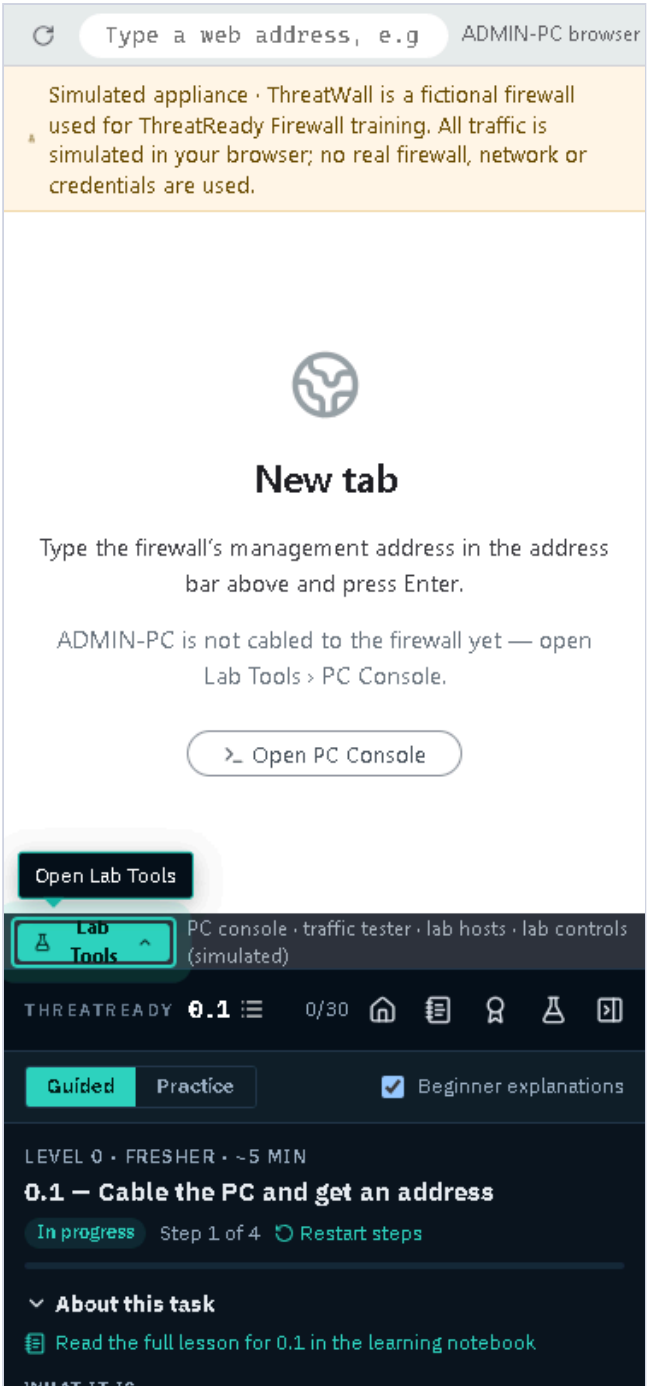
flow-devsecops-lab0-375x812.png

flow-redteam-tab-terminal-375x812.png



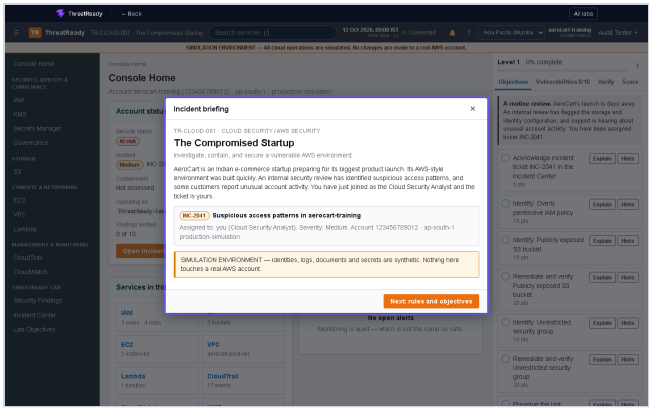
flow-soc-case-after-4s-1366x768.png



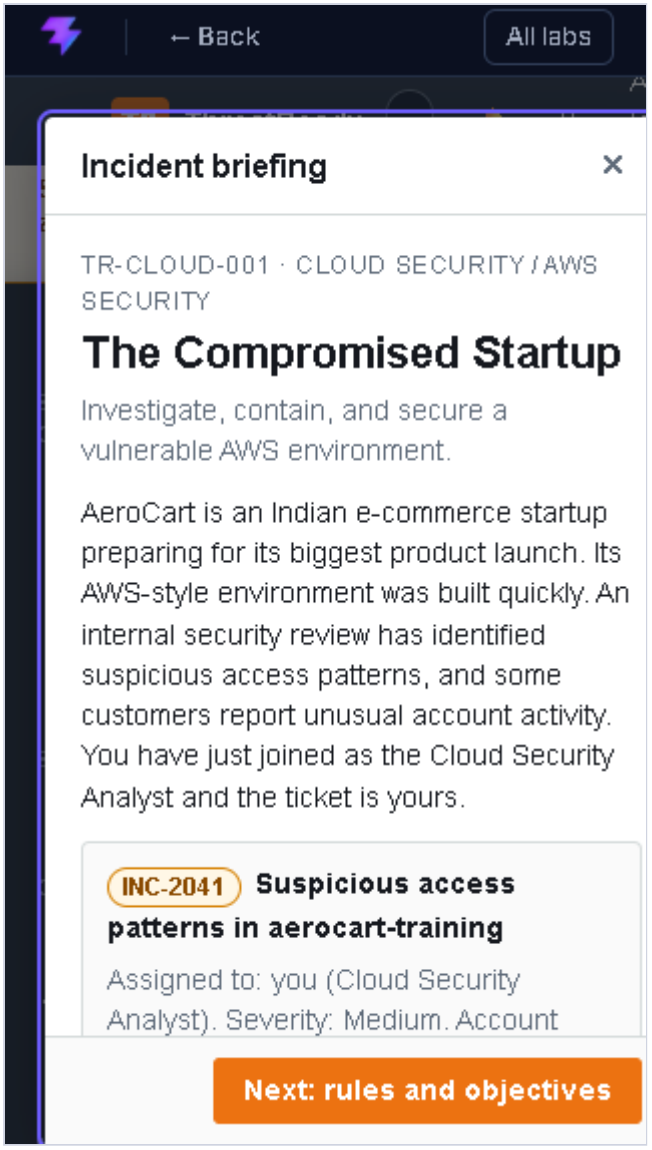


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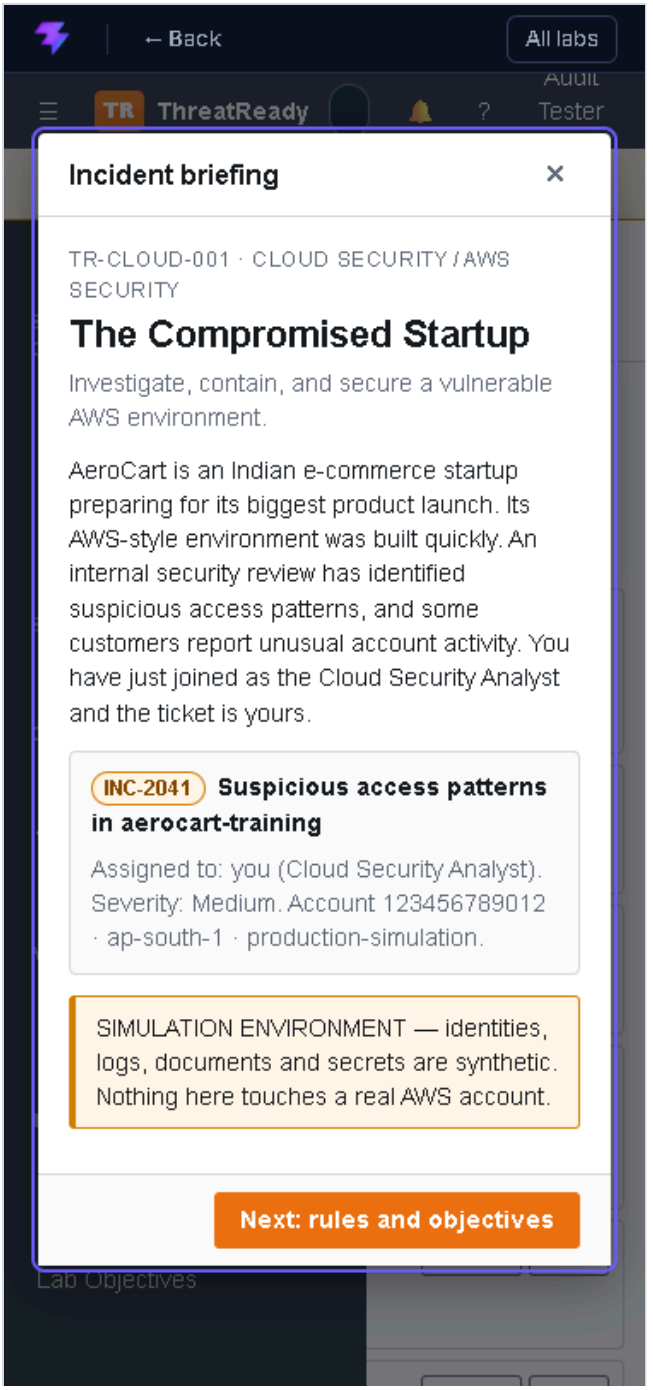
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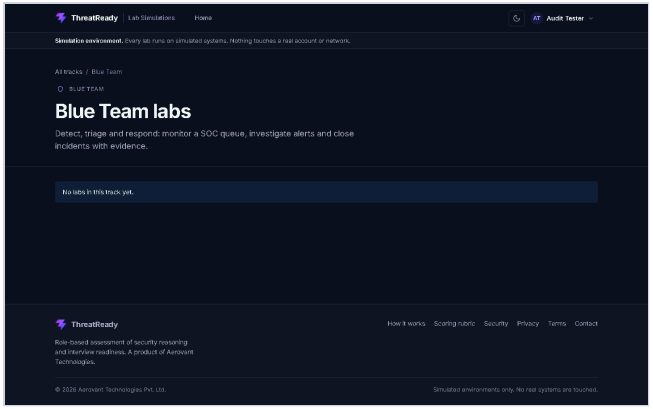


labs-console-1440x900.png

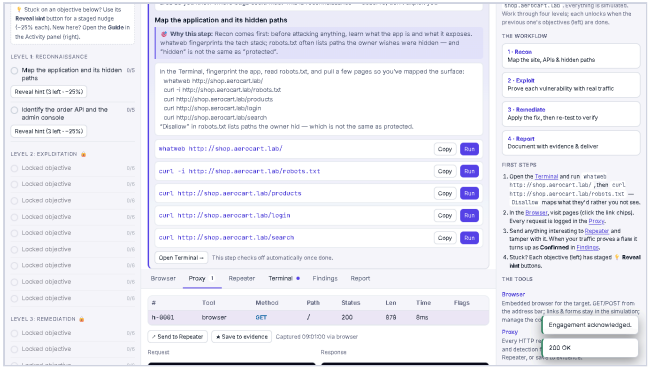


labs-console-320x568.png

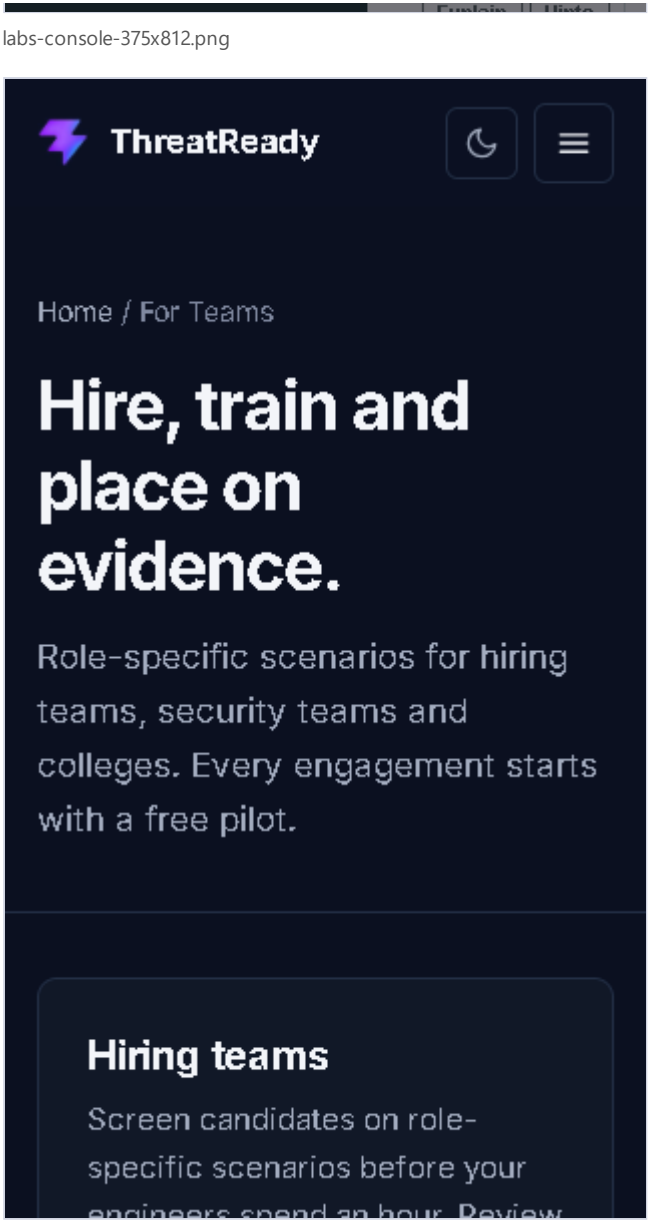




labs-track-blue-1440x900.png



light-redteam-proxy.png



site-pilot-320x568.png

This is a **sample report** with a fictional user — the exact format you receive after each session. Start your first session to generate yours.

ThreatReady

SAMPLE SKILL REPORT

See exactly what hiring managers receive

Fictional user · Real format

This is the exact report you get after each ThreatReady session — the candidate's full transcript, side-by-side with the ideal answer, scores per question, the reasoning behind every deduction, and a skills chart you can use in real hiring decisions. Scroll down to explore.

- Full Q&A transcript
- Skills chart
- Reasoning per deduction
- Integrity audit included

Home

Guided practice, sample scenario, simulated data

Newbie Demo Mode

Learn how a SOC analyst investigates security events, reads logs, collects evidence, validates findings, and completes an incident through a guided, step-by-step demonstration.

Demo ticket analyst **dev.analyst**

Before you start

- A Security Operations Centre (SOC) watche an organisation's logs and reacts when something looks wrong.
- The work is split into three tiers. Tier 1 (L1) looks at every alert and decides whether it i real. Tier 2 (L2) works out exactly what happened. Tier 3 (L3) stops the attack and cleans up. In this demo you play all three, o after the other, on the same ticket.
- You will work one sample alert from the first

soc-demo-320x568.png

AEROVANT TECHNOLOGIES THREATREADY Learning notebook

ThreatReady Firewall

ThreatReady Firewall by Aerovant Technologies: hands-on firewall administration training from first cable to incident containment. You configure a fictional firewall, ThreatWall, in a lab that runs entirely in your browser: cabling, interfaces, rules, VPNs, high availability and a final incident.

Learner name — Shown on your training record

CONTINUE WHERE YOU LEFT OFF

Task 0.1 — Cable the PC and get an address

Level 0 — Access · ~5 min · step 1 of 4

Continue →

LEARNING NOTEBOOK

Every task explained, with flow diagrams

What each feature really is, what happens inside the firewall, why every step matters, common mistakes and self-check questions. Read it here or download it as a PDF.

Read online | Download PDF

site-sample-report-375x812.png

tw-home-375x812.png


AEROVANT TECHNOLOGIES
THREATREADY
Home
Training record
Learner name not set · ThreatReady Firewall · generated 10/23/2026, 11:12:50 AM · every check re-run against the current lab configuration
In progress: 0 of 30 tasks verified
Print or save as PDF
Download my record (JSON)
SUMMARY
0/30
tasks verified now
0
failed attempts recorded
0
need re-verify
—
first to last completed task
Guided
mode at record time
Down
HQ↔Branch tunnel

tw-record-375x812.png